



CNN-attention combined with improved transformer model for medium- and long-term SST prediction

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ABSTRACT

Accurate prediction of sea surface temperature (SST) is essential for understanding climate change patterns, protecting marine ecology, and warning of natural disasters. It has been an important research topic in the fields of ocean and atmosphere. In recent years, deep learning has been increasingly applied to SST prediction. However, research on medium- to long-term global SST predictions remains insufficient. Existing prediction methods are mainly based on Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and their variant architectures, which exhibit insufficient capability to extract long-distance spatiotemporal teleconnection features. Meanwhile, prediction models integrating Transformers also face problems such as weak local feature fusion ability, limited prediction accuracy, and high model complexity. To address these issues, we propose an SST prediction method based on CNN-Attention combined with an improved Transformer (CNN-AT Transformer). Firstly, a CNN combined with an attention mechanism is used to extract and fuse the shallow semantic features of oceanic and atmospheric factors. Then, by embedding a convolutional layer into the Transformer sublayer to extract and fuse the local feature information. Additionally, a method is proposed to gradually transform global attention into local attention within the window by progressively decreasing the patch size in stages, thereby reducing computational complexity and minimizing prediction errors. A series of experimental results on real datasets show that our method has strong spatiotemporal relationship mining ability, and is significantly better than the baseline models in short-medium and long-term SST predictions.

1. Introduction

The ocean is not only the cradle of life, but also an important part of the global atmospheric system, which is crucial for maintaining the ecological balance and climate stability of the Earth. Sea surface temperature (SST) is an important covariate for the exchange of heat, power, and water vapor between the sea and air. It is also an important indicator for the study of the marine environment and climate change, which has a significant impact on global climate change, marine ecological protection, marine fisheries, and transportation. Therefore, accurate prediction of SST is crucial for understanding climate change, protecting marine ecology, warning of natural disasters, and has been one of the important research topics in atmospheric and ocean-related fields.

Currently, SST prediction methods are mainly categorized into Numerical Weather Prediction (NWP) methods and data-driven methods. NWP methods are still the mainstream methods, such as National Centers for Environmental Prediction-Global Forecast System (NCEP-GFS),

European Centre for Medium-Range Weather Forecasts High-Resolution Ensemble Prediction System (ECMWF-HRES) (Heming et al., 2019), and China's Global/Regional Assimilation and Prediction System (GRAPES) (Yu et al., 2022), etc. NWP methods are primarily based on the principles of fluid dynamics and thermodynamics, and they predict the SST for a future period by constructing mathematical equations of fluid motion for the atmosphere, ocean, and dynamic environmental systems. However, NWP methods have a huge computational cost and require a long time for forecast processing. The data-driven methods are to mine the variation rules of SST through a large amount of historical data, so as to predict the changes of SST, which are mainly divided into traditional statistical learning methods (Neetu et al., 2011; Ying et al., 2019), machine learning-based methods, and deep learning-based methods. Machine learning-based methods include SVM-based methods (Lins et al., 2013; Qi et al., 2019), Random Forest-based methods (Otsuka et al., 2018), Bayesian network-based methods (Bounceur et al., 2020), integrated learning methods (Wolff et al., 2020), and etc. These methods

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usually predict the SST of a single observation, and their models are relatively simple with weak feature extraction capability and low prediction accuracy.

In recent years, with the rapid development of deep learning technology, researchers have begun to apply deep learning methods to marine data analysis and prediction. Currently, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and their variant models are widely used in SST prediction. Yang et al. (2017) build a CFCC-LSTM model to predict SST. It uses RNN to capture temporal features and integrates them into CNN to calculate the spatial correlation between points in the region. Hao et al. (2023) employed ConvLSTM and ST-ConvLSTM networks to predict the sea surface temperature (SST) in the South China Sea. Pan et al. (2024) stacked multiple ConvLSTM layers to extract spatiotemporal features and incorporated an attention mechanism to weight historical features, enabling daily and monthly global SST predictions. Azhary and Minaoui (2025) proposed a ConvLSTM model based on dual attention mechanisms, utilizing convolution to capture spatial dependencies and the LSTM architecture to model temporal sequences, while integrating dual attention mechanisms to predict SST along the Moroccan coastline. Chen et al. (2025) introduced the SVRNN model, which integrates a bidirectional state-space processing mechanism, a decoupled memory module, and an LSTM architecture. This model effectively captures the complex spatiotemporal dependencies in SST data, improving prediction accuracy. Bilgili et al. (2025) utilized the Gated Recurrent Unit (GRU) model, Long Short-Term Memory (LSTM) neural network techniques, and the Seasonal Autoregressive Integrated Moving Average (SARIMA) statistical model to study global monthly SST temperature variations from 2023 to 2050. These models possess strong spatiotemporal feature extraction capabilities. However, they still exhibit limitations in capturing long-distance and spatiotemporal teleconnection features, and their computational efficiency is inferior compared to the models based on transformers.

Since the advent of the Transformer model, its excellent performance in the field of natural language processing has prompted researchers to explore its potential application in the field of SST prediction. Hitawe et al. (2024) proposed a new stacking architecture called StackPred based on Transformer, which improves the performance of a single Transformer model by capturing temporal dependencies and modeling sequence data to enhance the accuracy of SST and Wind Speed (WS) predictions. Chen et al. (2024) proposed TempProNet which fuses Transformer with CNN and adopts a deep-learning end-to-end codec network architecture to efficiently predict the temperature of the entire seawater profile in one go. Song et al. (2024) proposed the STVformer model. By integrating auxiliary variables such as short-wave radiation through the heat budget equation, they designed a multivariate feature representation module and a spatiotemporal variable saliency attention mechanism, which can effectively capture the multi-factor coupling relationship and long-term spatiotemporal dependence of sea surface temperature. Jia et al. (2024) proposed TL-iTransformer, which combines the iTransformer architecture with transfer learning. This model pretrains on rich ocean data and effectively solves the problem of sparse data in the target sea area through adaptive fine-tuning. Wu et al. (2024) proposed a Transformer model combined with physical information, namely PGTransNet, for predicting the three-dimensional ocean temperature and salinity in the tropical Pacific. This model designs the loss function by integrating the thermodynamic equation and embeds the PDO (Pacific Decadal Oscillation) and NPGO (North Pacific Gyre Oscillation) indices to capture long-term trends. Experiments show that the model is more accurate in coastal areas. Dai et al. (2024) developed the TransDtSt-Part model by embedding temporal information in the Transformer, combining attentional distillation and partial stacked connectivity, and using generative decoding to predict longer-term daily sea surface temperature (SST). Qi He (He et al., 2025) proposed a multi-scale periodic Transformer (MSPT) model. This model uses the Fast Fourier Transform (FFT) to decompose different periodic scales and employs the Transformer to learn the temporal variation characteristics on dif-

ferent time scales. Moreover, it uses a multi-periodic scale divider and a cross-scale aggregator to perform weighted aggregation on features. Experiments on the South China Sea data have verified that the model has high prediction accuracy. Song et al. (2025) proposed a China coastal sea surface temperature (SST) prediction model based on global-local spatiotemporal information interaction guided fusion. This model uses self-attention and cross-attention to solve the early interaction of spatiotemporal information, and adopts the GL-LSTM-Transformer module to realize the interaction and fusion of feature information with different spatiotemporal receptive fields. Experiments on the Bohai Sea, East China Sea, and South China Sea datasets have verified its effectiveness.

Transformer-based methods have certain advantages in improving the accuracy of SST prediction. However, existing methods still face problems such as weak spatiotemporal feature extraction and fusion capabilities, limited prediction accuracy, and high model complexity. For this reason, we propose a sea surface temperature prediction model based on the combination of CNN-Attention and an improved transformer. This model is based on atmospheric and oceanic elements. It uses CNN combined with an attention mechanism to extract shallow semantic features, enhances the local feature extraction ability through an improved Transformer, and adopts a method of gradually reducing the block size in stages to reduce the computational complexity. Thus, it realizes the prediction of global sea surface temperature and achieves high prediction accuracy. The main contributions of this paper are as follows:

1. A SST prediction model based on CNN-Attention combined with an improved Transformer (CNN-AT Transformer) is proposed. The model fully utilizes the spatial feature extraction and fusion capabilities of the CNN and Attention mechanisms, as well as the long time-series feature extraction capability of the improved Transformer model, and realizes high-precision prediction of SST.
2. A shallow feature extraction and fusion module is designed to extract semantic features from the spatiotemporal sequences of SST and sea level pressure data using channel-wise convolution. This helps capture broader spatiotemporal dependencies such as spatiotemporal teleconnections, improving the model's prediction accuracy.
3. An improved Transformer model is proposed, which embeds CNN convolutional layers into Transformer sublayers to extract and fuse local feature information, addressing the issue of boundary artifacts between predicted image patches in the traditional Transformer model. Meanwhile, a method of gradually reducing the patch size in stages is proposed, which gradually shifts the global attention to local attention within the window, reducing the computational complexity and prediction error.
4. A series of experiments are conducted on our model using the NOAA ERSST V5, NOAA OISST V2 and NCEP/DOE Reanalysis II datasets, as well as the datasets from CMIP5 and CMIP6. The experimental results show that our model has high prediction performance and is significantly better than the baseline model.

The rest of the paper is organized as follows: Section 2 discusses the related work on SST prediction, and Section 3 details the framework of the proposed methodology and implementation details. Experimental results and in-depth discussions are given in Section 4. Finally, Section 5 summarizes the paper.

2. Related work

The methods for predicting SST are mainly divided into two categories: NWP methods and data-driven methods. NWP methods are primarily based on the principles of ocean physics and the coupling between the atmosphere and the ocean (Jiaying et al., 2020; Yajuan et al., 2022). They establish complex mathematical equations and conduct simulations through high-resolution grids. These methods have a solid physical foundation and require the solution of complex fluid dynamics and thermodynamics equations. However, they entail high computational costs and rely on parameterization schemes. Data-driven methods

combine a large amount of data to search for statistical patterns for prediction. They are divided into traditional statistical learning methods, machine learning-based methods, and deep learning-based methods that have emerged in recent years.

In data-driven methods, traditional statistical learning methods include Empirical Orthogonal Function (EOF) (Neetu et al., 2011), Linear Regression and Auto Regressive Integrated Moving Average (ARIMA) (Ying et al., 2019), and so on. Machine learning-based methods mainly include SVM (Lins et al., 2013; Qi et al., 2019), Random Forest (Otsuka et al., 2018), Bayesian methods (Bounceur et al., 2020) and Model Integration (Wolff et al., 2020). These methods are mainly used to predict the SST at a single observation point or within a single sea area. Since they cannot extract deep-level features from the SST data and require manual feature selection, their prediction accuracy is relatively low. In recent years, they have gradually been replaced by the deep learning-based methods.

Deep learning-based methods are mainly categorized into Feed-forward Neural Network (FNN)-based, Convolutional Neural Network (CNN)-based, and Recurrent Neural Network (RNN)-based methods. The RNN-based methods usually combined with CNN or Graph Neural Networks (GNNs) to assist the extraction of spatial features. Scholars at home and abroad have long applied FNN to the SST prediction problem (Jiaying, Jiye and Jingjia, 2020; Yajuan, Zhenya, Meng, Qi, Ying and Fangli, 2022; Bollivier, Eifler and Thiria, 2000; Wu, Hsieh and Tang, 2006; Jiakang, Qijie, Ying and Honglin, 2017; Hossein, Valentina and Steven, 2018; Wei, Guan and Qu, 2020; Wei, Guan, Qu and Guo, 2020). FNN-based methods are characterized by simplicity and ease of training, etc. However, the FNN structure is too simple to effectively capture spatial-temporal variation features, and has been gradually replaced by CNN- and RNN-based methods. CNNs are generally combined with RNNs to extract spatial-temporal features from SST data. In this regard, Rice et al. (2020) implemented SST prediction by combining Koopman and CNN, which provides some physical interpretability. Feng et al. (2021) used a Temporal Convolutional Network (TCN) model to predict the long-term SST in the Indian Ocean waters, and used a variety of oceanic and atmospheric factors affecting the SST as model inputs, which improved the prediction accuracy. RNN-based methods usually use the long-term sequence feature extraction capability of RNNs to model the variation patterns of SST (Zhang et al., 2017; Liu et al., 2019; Chen and Dong, 2019; Xie et al., 2020). Due to the neglect of the spatial characteristics of SST data, the prediction precision of these methods is relatively low. Therefore, subsequent studies have proposed methods and structures that combine temporal features with spatial features. Yang et al. (2017) proposed the CFCC-LSTM model, which incorporates convolutional layers into a fully connected LSTM network for SST prediction, effectively leveraging both temporal and spatial dependencies in the data. Similarly, Xuewei and Zhen (2022) combined CNN and RNN models to simultaneously learn the spatiotemporal dependencies in the data. Qiao et al. (2023) combined the XGBoost model, PredRNN network, and attention mechanism to propose a new ensemble learning method for SST field prediction. The effectiveness of this method was verified on the Bohai Sea and North Sea datasets. Bai et al. (2025) proposed a multi-scale spatiotemporal attention network called MUSTAN. It employs a self-attention mechanism to prioritize relevant temporal influences at specific scales and combines window-based shifted self-attention with each scale to emphasize information-rich spatial components. Experimental results demonstrate that MUSTAN achieves high SST prediction accuracy.

In addition, graph neural networks have also been applied to the prediction of sea surface temperature. Sun et al. (2021) combines graph neural networks and RNNs. Using graph neural networks to learn the dependency relationships among different points on the sea surface is also a method for extracting spatial features. Gao et al. (2023) proposed a Global Spatiotemporal Graph Attention Network (GSTGAT). The Global Graph Attention Network (GGAT) is used to capture the global dynamic spatial correlations of nodes, and the Gated Temporal Convolutional

Network (GTCN) module is used to capture nonlinear temporal correlations. Its performance has been verified on the Bohai Sea and South China Sea datasets. Xiao et al. (2024) proposed a Graph Spatiotemporal Sampling Aggregation (GSSA) method. This method mainly consists of two parts: spatial information aggregation and temporal information convolution. GSSA endows the model with inductive ability through neighbor sampling, captures spatial details through spatial information aggregation, and grasps temporal dynamics through temporal information convolution, thereby improving prediction accuracy. To facilitate understanding, we provide a concise summary of data-driven SST prediction methods in the literature from three aspects: prediction region, forecasting horizon, and utilized datasets, as detailed in Table 1.

From the above analysis, it can be seen that the existing deep learning-based SST prediction methods are mainly aimed at SST prediction of small sea areas or individual points, and have insufficient ability to extract long-range and spatiotemporal teleconnection features, resulting in poor accuracy of medium- and long-term SST prediction. This is the motivation of our work.

3. Methodology

3.1. Description of the problem

SST prediction involves forecasting future SST sequences in a specific ocean region based on historical SST observations. Typically, we use a two-dimensional grid of $H \times W$ to represent a sea area, where H is the number of rows and W is the number of columns. Each cell represents an observation point, and its value represents the observed value at observation point. Thus, the SST of this sea area at time t can be represented by a three-dimensional tensor $X_t \in \mathbb{R}^{H \times W \times C}$, where C is the number of temperature-related elements. In this work, we use two elements, SST and Sea Level Pressure (SLP), to predict the SST, so the value of C is 2. Now, the SST prediction problem can be defined as: given the observed sequence of SST over the past n time steps $X_{t-n+1}, X_{t-n+2}, \dots, X_t$ ($X_i \in \mathbb{R}^{H \times W \times 2}, i \in \{t-n+1, t-n+2, \dots, t\}$) and establish a model to compute the most probable set of values as the future SST sequence for the next k time steps $\hat{Y}_{t+1}, \hat{Y}_{t+2}, \dots, \hat{Y}_{t+k}$ ($Y_{t+i} \in \mathbb{R}^{H \times W \times 1}, i \in \{1, 2, \dots, k\}$), which can be described by Eq. (1).

$$\hat{Y}_{t+1}, \hat{Y}_{t+2}, \dots, \hat{Y}_{t+k} = \underset{Y_{t+1}, \dots, Y_{t+k}}{\operatorname{argmax}} P(Y_{t+1}, \dots, Y_{t+k} | X_{t-n+1}, \dots, X_t; \theta) \quad (1)$$

Where θ is the model parameters and the model can be any machine learning model, such as CNN, RNN, Transformer, etc.

3.2. The framework of the model

For the SST prediction problem described in Section 3.1, we propose CNN-AT Transformer, a new SST prediction model based on CNN-Attention combined with the improved Transformer, and its overall framework is shown in Fig. 1. The CNN-AT Transformer model is composed of three main components: the shallow feature extraction and fusion module, the spatiotemporal Transformer encoder and the multi-output SST prediction decoder. The shallow feature extraction and fusion module initially extracts shallow features from SST and sea level pressure data, and then employs an attention mechanism to fuse the extracted features, achieving better prediction accuracy and interpretability. The spatiotemporal Transformer encoder module is mainly used to further extract and fuse the spatiotemporal dependency features between SST and sea level pressure data. We improve the traditional Transformer model by embedding a CNN convolutional layer in its sublayer to mine and fuse the local feature information, which solves the defects of the traditional Transformer model. Meanwhile, we adopt a phased approach to gradually reduce the patch size, progressively transitioning global attention to local attention within windows. This reduces the model's computational complexity and improves prediction accuracy. The multi-output SST prediction decoder module decodes and predicts

Table 1

A brief comparison of the data-driven SST prediction methods in the literature.

Model	Prediction area	Prediction horizon	Data used
GA + EOF [Neetu et al. (2011)]	Arabian Sea	2–4 weeks	SST
EEMD + ARIMA [Ying et al. (2019)]	the South China Sea	1–7 days	SST
SparkDTW + SVM [Qi et al. (2019)]	–	5 days	SST
SVM [Otsuka et al. (2018)]	Gokasho Bay, Matoya Bay, Ago Bay	1 day	SST, Wind Speed
BSTS [Bounceur et al. (2020)]	Red Sea	1–5 months	Regional SST, Climate metrics
MLM Ensemble [Wolff et al. (2020)]	Global	Daily	Atmospheric Data, SST
CFCC-LSTM [Yang et al. (2017)]	China Ocean, Bohai Sea	1, 7, 30 days	SST
ST-ConvLSTM [Hao et al. (2023)]	South China Sea	1 day	SST
EAM [Pan et al. (2024)]	Global	Daily: 15 days; Monthly: 12 months	Global SST
EDDA-ConvLSTM [Azhary and Minaoui (2025)]	Moroccan coast (21 °N–36 °N, 6 °W–19 °W)	8 weeks	NOAA OISST v2
SVRNN [Chen et al. (2025)]	Taiwan Strait	12 h	SST
StackPred [Hittawe et al. (2024)]	Red Sea	1–5 months	SST, Wind Speed
TemproNet [Chen et al. (2024)]	South China Sea	everyday	SST, Sea Level Anomaly, Sea Surface Wind
STVformer [Song et al. (2024)]	East China Sea, South China Sea	7 days	SWR, LWR, LHF, SHF, SST
TL-iTransformer [Jia et al. (2024)]	Coastal waters of British Columbia, Canada (12 light-houses)	Multiple prediction lengths	SST
PGTransNet [Wu et al. (2024)]	3D temperature and salinity in the tropical Pacific	Multi-step prediction; evaluated on accuracy and physical consistency	3D SST, Salinity, PDO, NPGO
TransDtSt-Part [Dai et al. (2024)]	China Seas (Bohai, Yellow Sea, East China Sea, Taiwan Strait, South China Sea)	30, 60, 90, 180, 270, 360 days	NOAA satellite SST data
MSPT [He et al. (2025)]	4 locations in the South China Sea	1–30 days	SST, Periodic Features
GL-ST [Song et al. (2025)]	Coastal waters of China (Bohai, East China Sea, South China Sea)	1, 3, 7, 10, 14days	global-local SST
FNN [Bollivier et al. (2000)]	Cape Blanc, Mauritania	1–8 days	SST, Wind Stress
FNN [Wu et al. (2006)]	Tropical Pacific Ocean	3–15 months	Sea Level Pressure, SST
FNN + CEEMD [Jiakang et al. (2017)]	Northeast Pacific, Equatorial East Central Pacific	—	SST
FNN [Wei et al. (2020)]	South China Sea	1 month	SST
LSTM [Wei et al. (2020), Zhang et al. (2017), Chen and Dong (2019)]	China's Sea Areas	1–12 months	SST
ConvConsKAE [Rice et al. (2020)]	Gulf of Mexico	1–180 days	SST
TCN [Feng et al. (2021)]	Indian Ocean	1 month	Oceanic and Atmospheric Data
MSH-LSTM [Liu et al. (2019)]	Equatorial Pacific	1–8 months	SST
GED + DIL [Xie et al. (2020)]	Bohai Sea, South China Sea	Daily: 1, 3, 7days Weekly: 1, 3weeks Monthly: 1 month	SST
ConvGRU [Xuewei and Zhen (2022)]	Western Pacific	Monthly: 1 month	NOAA OISST v2
ELA-PredRNN-AT [Qiao et al. (2023)]	Bohai Sea, South China Sea	1, 3, 7days	SST
MUSTAN [Bai et al. (2025)]	Bohai Sea, South China Sea, Yellow Sea	1, 3, 7 days	SST
TSGN [Sun et al. (2021)]	Northwest Pacific	1, 3, 7 days	SST
GSTGAT [Gao et al. (2023)]	Bohai Sea, South China Sea	Daily: 1, 3, 7days Weekly: 1, 3weeks Monthly: 1 month	SST
GSSA [Xiao et al. (2024)]	Bohai Sea, South China Sea	Daily: 1, 3, 7days Weekly: 1, 3weeks Monthly: 1 month	SST

the future SST based on the feature representation extracted by the spatiotemporal Transformer encoder.

3.3. Shallow feature extraction and fusion module

The problem description shows that SST observations at each time step can be treated as multi-channel images. Directly dividing these images into patches and feeding them into the improved Transformer model would hinder information exchange between patches, thereby degrading prediction accuracy. To solve this, we design a sub-network consisting of CNN and attention mechanism for shallow feature extraction and fusion, whose structure is shown in Fig. 2. The network extracts features from different SST and sea level pressure data through channel-

by-channel convolution, and adaptively learns the importance of different features and fuses them based on the channel-attention mechanism, which yields a better feature representation and enhances the interpretability of the model.

First, the feature extraction is performed on the original input X using a convolutional network as shown in Eq. (2).

$$F_{\text{extract}}: X \rightarrow U, X \in \mathbb{R}^{H \times W \times C}, U \in \mathbb{R}^{H \times W \times rC} \quad (2)$$

Where U is the feature representation, r is the number of convolution kernels, and the convolution process is shown in Eq. (3).

$$u_{c,k} = v_{c,k} * x_c, c \in \{1, 2, \dots, C\}, k \in \{1, 2, \dots, r\} \quad (3)$$

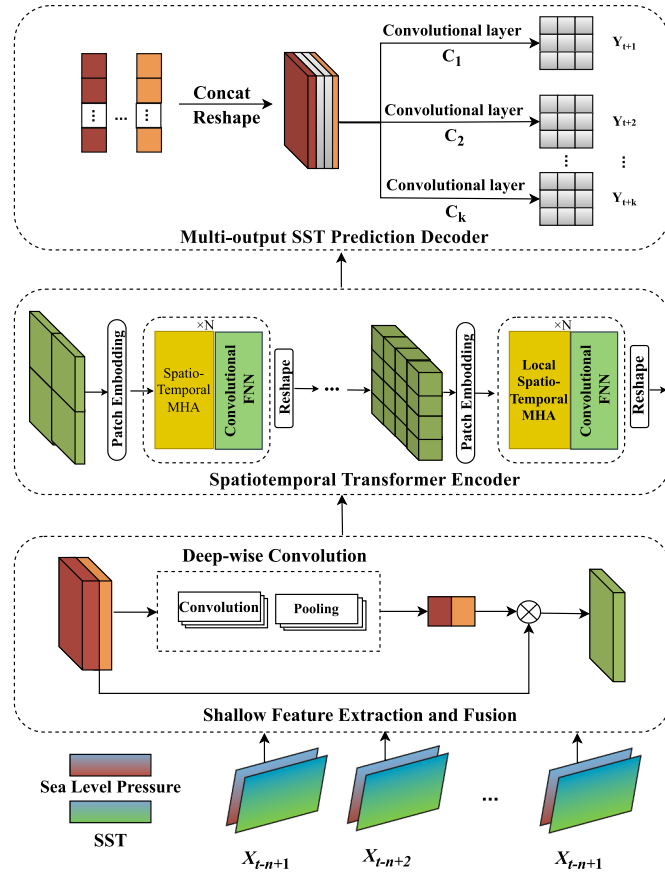


Fig. 1. The framework of our SST prediction model.

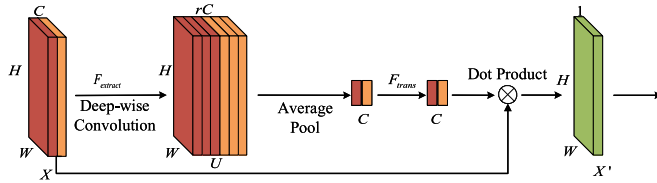


Fig. 2. Structure of shallow feature extraction and multi-source fusion module.

Where x_c denotes the c -th channel of the input features, $v_{c,k}$ denotes the k -th convolution kernel of the c -th channel, $u_{c,k}$ denotes the feature representation generated by the k -th convolutional kernel of the c -th channel, and $*$ denotes the convolution operation.

Then, global pooling is applied to the convolution output results for feature compression, as shown in Eq. (4).

$$z_c = \frac{1}{r \times H \times W} \sum_{k=1}^r \sum_{i=1}^H \sum_{j=1}^W u_{c,k}(i, j) \quad (4)$$

Finally, the model adaptively adjusts the compressed feature vector, as shown in Eq. (5).

$$S = F_{trans}(z; W, b) = \sigma(\text{ReLU}(zW_1 + b_1)W_2 + b_2) \quad (5)$$

$$W_1 \in \mathbb{R}^{c \times c'}, W_2 \in \mathbb{R}^{c' \times c}, b_1 \in \mathbb{R}^{c'}, b_2 \in \mathbb{R}^c$$

Where W and b denote the weight and bias of the fully connected layer, respectively. The feature vector $S = (s_1, s_2, \dots, s_c)$, each value represents the importance of the corresponding feature map.

Finally, the feature vector S is multiplied with the original feature map to take the average value, and the final fused single-channel feature

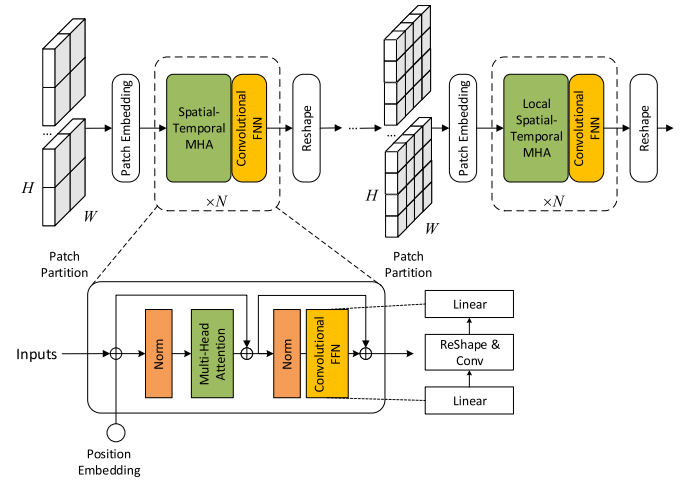


Fig. 3. The structure of spatiotemporal transformer encoder.

map is obtained, as shown in Eq. (6).

$$X'(i, j) = \sum_c x_c(i, j) \cdot s_c \quad (6)$$

3.4. Spatiotemporal transformer encoder

We propose the spatiotemporal Transformer encoder to further extract spatiotemporal dependencies from the fused feature representation. Its structure is shown in Fig. 3. The encoder consists of multiple stages, each of which consists of three parts: patch embedding, encoding, and reshape. Patch embedding is responsible for partitioning feature maps and mapping them into one-dimensional vector representations. Our experiments show that using larger fixed patches can reduce the prediction details, while smaller patches lead to long sequences that are difficult to optimize. Therefore, we propose a method that gradually reduces the patch size. In the first stage, larger patches are used to allow the model to compute attention among the larger patches. In the subsequent stages we gradually reduce the patch size and use a non-overlapping window to partition the patches so that the model can focus more on the local information and reduce the complexity of global attention. We implement encoding and promote information fusion by embedding a Convolutional FNN sub-layer in the original Transformer encoder. Convolutional FNN will be described in detail in Section 3.5.

As shown in Fig. 3, we adopt a progressively reducing patch size strategy, which requires patch division and local self-attention computation for many times. Therefore, the position relationships between patches are particularly important. If position encoding is not added, it would cause confusion in semantic information, affecting the model's sequence modeling capability. Therefore, we adopt the strategy of learnable position encoding combined with fixed absolute position encoding. Specifically, in the calculation of global attention, we use learnable absolute position encoding, which is commonly used in BERT model, to allow the model to adaptively learn complex spatiotemporal distance relationships. At the same time, after patch division, the 1D fixed absolute position encoding in the original Transformer is added to the patches inside each local window respectively, so that the attention mechanism can obtain the local position relationship. The calculation method of the fixed absolute position encoding is shown in Eq. (7).

$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d}}\right) \quad (7)$$

$$PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

Where pos denotes the position of the token in the sequence, i denotes the i -th dimension of the positional encoding vector of the token, and d denotes the length of the token vector, which may vary across layers in the model.

Let the input to each layer be $X' \in \mathbb{R}^{T \times H \times W \times C}$, assuming that N in Fig. 3 is 1, and disregarding residual connections. The patch partition module first divides the input feature maps into 3D patches, and then the patch embedding module performs embedding and maps it into a one-dimensional vector. The operations of these two modules can be represented by a transformation $F_{trans} = X' \rightarrow I, I \in \mathbb{R}^{T' \times H' \times W' \times D}$. The positional encoding is subsequently added to the one-dimensional vector formed by patch embedding to form the corresponding embedded feature representation. Next, for the embedded feature representation, the multi-head attention mechanism is used to calculate the attention scores for each head separately, and the output results are obtained. The calculation method is shown in Eq. (8).

$$head_i = \text{Attention}(Q_i, K_i, V_i) = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i$$

$$Q_i = I W_i^Q, K_i = I W_i^K, V_i = I W_i^V$$

$$W_i^Q \in \mathbb{R}^{D \times d_q}, W_i^K \in \mathbb{R}^{D \times d_k}, W_i^V \in \mathbb{R}^{D \times d_v} \quad (8)$$

Where $head_i$ represents the output of the i -th attention head, d_q, d_k and d_v represent the dimensions of Q, K and V in each head respectively, and $d_q = d_k = d_v$ is set to simplify the modeling. Then concatenate the outputs of each head and perform linear transformation to form a new feature representation, see Eq. (9).

$$H = F_{out}(head_1, head_2, \dots, head_n)$$

$$= \text{Concat}(head_1, head_2, \dots, head_n) W, W \in \mathbb{R}^{n d_v \times D} \quad (9)$$

To learn local features within each patch and relationships between neighboring patches, the model uses a Convolutional FNN network to nonlinearly transform and fuse the outputs of multiple attention layers. The network contains two fully connected layers and one convolutional layer. The operation of the Convolutional FNN is described in Eq. (10).

$$H_1 = \text{Reshape}(\text{GeLU}(H W_1 + b_1))$$

$$H' = \text{Reshape}(\text{Conv}(H_1)) W_2 + b_2 \quad (10)$$

Where $W_1 \in \mathbb{R}^{D \times q d^2}, W_2 \in \mathbb{R}^{q d^2 \times D}, q$ is the number of 2D patches contained in each 3D patch and GeLU is an activation function.

Finally the feature map structure converted to a sequence structure needs to be reduced, i.e., the inverse transformation of $F_{trans}, F_{anti-trans}: H' \rightarrow O, O \in \mathbb{R}^{T \times H \times W \times C}$. The Reshape module is used to reorganize the data for the next stage of operation.

3.5. Convolutional FNN module

We found through experiments that if we follow the FNN design in traditional Transformer, there are obvious problems in the decoding prediction process. Specifically, there are obvious demarcation lines between patches in the predicted SST views, especially in cases when the patches are large. This is due to the loss of long-range distance relationships between patches in the self-attention mechanism, and the points on the boundaries fail to learn information on the boundary of neighboring patches. Additionally, Transformer models struggle to directly utilize information from nearby points of a patch during training, leading to this phenomenon. To address this, we introduced this prior knowledge into the model design, enabling the model to learn and utilize information from neighboring patches. Convolutional layers are suitable structures for integrating local information, so we attempted to incorporate them into the Transformer model. Although there has been quite a bit of work combining convolution and Transformer (Xu et al., 2021), the migration of these models and methods to SST data has been poor.

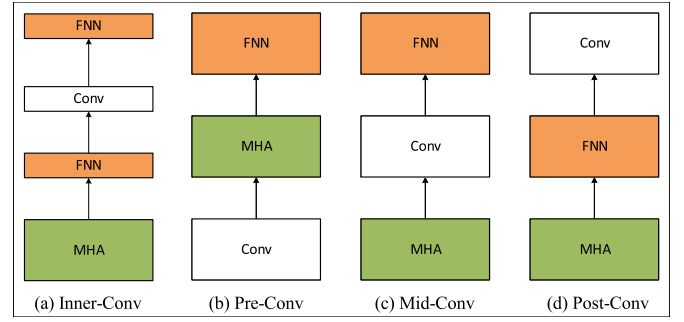


Fig. 4. Schematic of the different ways convolution can be added to the transformer.

Therefore, we used experiments to study the effects of adding convolutional layer at different positions in each layer of Transformer model, as shown in Fig. 4.

We found through the experiments that the latter three are very poor, although the convergence is faster in the early stage, but the final model convergence is worse than the original model, while the first method of inserting the convolutional layer between the two linear layers of the FNN sublayer works optimally. Specifically, the first linear layer in the FNN sublayer is used to learn the features inside the patch, as shown in Fig. 4(a). Then the segmented patch is restored to a 2D feature map, or a 3D feature map if it is delineated in 3D, and then the second linear layer is used to restore the dimensionality of the patch by fusing the information between neighboring patches and re-segmenting the patch using convolution. In this way, each patch contains not only the local information of the corresponding location but also the information of its neighbors. At the same time, as the network level deepens, the receptive field of the deep convolutional kernel gradually increases, and due to the existence of residual connections (He et al., 2016), the Convolutional FNN constitutes an information flow path that can also gradually learn global information.

3.6. Multi-output SST prediction decoder

We present an innovative multi-output prediction decoder designed to accurately predict SST over multiple future steps based on the eigenstates extracted by the spatiotemporal Transformer encoder. The design of this decoder centers on the utilization of multiple convolutional layers in order to simultaneously predict the SST over multiple future steps. First, the input spatiotemporal data are feature extracted by a spatiotemporal Transformer encoder to obtain a series of feature representations $O = \{O_1, O_2, \dots, O_T\}$, where T denotes the time step, which contain rich spatiotemporal information and provide a basis for the subsequent prediction. These feature representations are then Concat and Reshape to form a shared feature representation S . The purpose of this step is to fuse features from different time steps to obtain a global feature representation that captures long-term dependencies, see Eq. (11).

$$S = \text{Reshape}(\text{Concat}(O)) \quad (11)$$

Where Concat denotes the feature splicing operation and Reshape denotes the reshaping of the Concatenated features to fit the input requirements of the subsequent convolutional layers. After obtaining the shared features S , we design multiple independent convolutional layers C_i , each of which is responsible for predicting the SST in the next steps. These convolutional layers work in parallel. For the i -th convolutional layer, the prediction process can be expressed as follows:

$$Y_i = C_i(S) \quad (12)$$

In order to improve the accuracy of the prediction, the outputs of all the convolutional layers are optimized using a joint loss function L , which can achieve a higher accuracy on long-period prediction than single-step prediction, providing a new and effective method for SST prediction.

3.7. Loss function

For SST prediction, MSE (Mean Squared Error) or MAE (Mean Absolute Error) loss function is usually used to optimize the neural network. However, MSE is more sensitive to the noise points, and in the absence of noise, when the data obeys normal distribution, MSE can theoretically guarantee the best prediction results. Since SST data do not obey normal distribution, using MSE as a loss function may not be optimal. Since the MAE loss function uses the absolute value of the difference between each prediction and the true target, it is not sensitive to noise points. At the same time, the gradient of MAE is not continuous and is non-differentiable at zero, which means its gradient calculation is more complex during backpropagation, reducing efficiency and resulting in optimization performance typically being inferior to MSE. Based on the above analyses, we combine the two for the loss calculation of SST prediction, moreover, since the model has multiple future prediction outputs, we directly equalize the loss of each moment, so that the final loss function for individual point prediction is shown in Eq. (13).

$$L(y, \hat{y}) = (1 - \mu)(y - \hat{y})^2 + \mu |y - \hat{y}| \quad (13)$$

where μ is the hyperparameter in order to balance the weights of the two loss functions.

4. Experiments

4.1. Datasets and preprocessing

For the medium- to long-term prediction task, we use the NOAA ERSST V5 (NOAA Extended Reconstructed SST V5) data to train, validate, and test the model, which is one of the datasets maintained by the National Oceanic and Atmospheric Administration (NOAA) to characterize SST variations, spanning the period from January 1854 to 2020. This dataset is obtained by statistically interpolating SST observations and provides monthly mean SST data with a latitude and longitude resolution of $2^\circ \times 2^\circ$ on a global scale. As for the sea level pressure data, we use the NCEP/DOE Reanalysis II dataset, which spans from 1979 to 2020, and since the time span is smaller than that of the SST data, the missing parts are replaced by the output of the historical simulation experiment of the climate model. All of the above data are available from the NOAA Physical Sciences Laboratory (PSL) website (<https://psl.noaa.gov/data/gridded/index.html>). To assess the model's short-term SST prediction ability, we used the NOAA OISST V2 dataset, which includes global high-resolution daily SST data (1981 to 2020).

We also use the outputs of historical climate simulation experiments (HCSEs) of several climate system models to pre-train the model. These data were obtained from the CMIP5 (Xiaoge et al., 2012), CMIP6 (Xiaoge et al., 2019) trials, which are the fifth and sixth phases of the Coupled Model Comparison Initiative (CMIP) organized by the World Climate Research Program (WCRP). These datasets can be downloaded from the Earth System Grid Federation (ESGF) website (<https://esgf-node.llnl.gov/projects/cmip5/>) and (<https://esgf-node.llnl.gov/projects/cmip6/>). The reference El Niño index is from the NOAA PSL laboratory in the United States. The detailed description of the datasets used for medium and long-term prediction of SST is shown in Table 2.

The short-term SST prediction dataset is divided in time as follows: Data from 2000 to 2016 are used as the training set, data from 2017 to 2018 are used as the validation set, and data from 2019 to 2020 are used as the test set.

4.1.1. Data preprocessing

Since the datasets come from different sources, there are subtle differences in their formats, resolutions, etc. Therefore, we have preprocessed the data to ensure its consistency. The data preprocessing mainly includes steps such as data alignment, missing value handling, edge padding, and standardization. The following is a detailed description.

Table 2

The description of data sets.

Data set	Description	Time range
training set	CMIP5, CMIP6 historical analog test output (Pre-training)	
	NOAA ERSST V5	
	NCEP/DOE Reanalysis II	1850–1970
validation set	NOAA ERSST V5	
	NCEP/DOE Reanalysis II	1971–1980
test set	NOAA ERSST V5	
	NCEP/DOE Reanalysis II	1981–2020

(1) Data Alignment. We unify the data spatial grid to $2^\circ \times 2^\circ$ by intercepting the latitude range and unifying the longitude, thereby achieving data alignment. For data with inconsistent resolutions, we adjust them through bilinear interpolation. This helps improve the modeling effect.

(2) Missing Value Handling. For the missing values in ERSST V5 caused by observational gaps, we use the spatiotemporal Kriging interpolation method to fill them. For the missing values in ERSST V5, we model the spatial and temporal correlations using the covariance function to fill them and correct the outliers. Regarding the systematic bias between CMIP model data and observational data, we use the Quantile Mapping method to calibrate and eliminate the distribution differences between the model output and the real observations.

(3) Edge Padding. To address the differences in the sizes of gridded data from different sources, we preprocess the data by padding 0 values at the edges. When calculating the loss and evaluating the model, the padded areas are not considered.

(4) Standardization. We standardize the input data to the interval $[0, 1]$. The calculation formula is shown in Formula (14).

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} - 0.5 \quad (14)$$

4.1.2. Evaluation metrics

In terms of model evaluation, we use three common metrics: Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and Structural Similarity Index (SSIM), to evaluate the SST prediction models. Their calculation formulas are shown in Eqs. (15)–(17). Among them, RMSE and MSE is used to evaluate the prediction effect of each independent observation point separately, while SSIM is used to evaluate the overall prediction effect of the entire sea region.

$$\text{RMSE}_l = \sqrt{\frac{1}{HWT} \sum_{t=1}^T \sum_{j=1}^W \sum_{i=1}^H (Y_t^{(i,j)} - P_{(t,l)}^{(i,j)})^2} \quad (15)$$

$$\text{MAE}_l = \frac{1}{HWT} \sum_{t=1}^T \sum_{j=1}^W \sum_{i=1}^H |Y_t^{(i,j)} - P_{(t,l)}^{(i,j)}| \quad (16)$$

$$\text{SSIM}_l = \frac{1}{T} \sum_{t=1}^T \frac{(2\mu_{Y_t} \mu_{P_{t,l}} + c_1)(2\sigma_{Y_t P_{t,l}} + c_2)}{(\mu_{Y_t} + \mu_{P_{t,l}} + c_1)(\sigma_{Y_t}^2 + \sigma_{P_{t,l}}^2 + c_2)} \quad (17)$$

Where l denotes the number of time steps ahead of the prediction, which is 1–24 months in the medium and long period prediction of SST, and 1–30 days in the short period prediction of SST, $P_{(t,l)}^{(i,j)}$ denotes the predicted SST at the time of t at the location of (i, j) , which is 1–30 days ahead of the predicted SST at the time of l , and $Y_t^{(i,j)}$ denotes the true SST at the time of t at the location of (i, j) . When evaluating the overall effect, the values of the indicators predicted by different steps ahead are calculated separately and then averaged.

4.2. Basic experimental analysis

This section sets the loss function and training optimization strategy for the medium and long-term SST prediction model through experiments, mainly including three groups of comparative experiments: loss

Table 3
Hyperparameter setting.

Parameters	Value
Epochs	30
learning rate	0.001
Weight Decay	0.05
β_1	0.9
β_2	0.999
DropPath	0.1
Patch Default Size	$4 \times 16 \times 16$
Default number of heads of long attention	4

function for SST prediction, optimal number of input months needed to predict the SST for the next 1–24 months, and SST prediction model pre-training experiments. The experiments in this section are all based on the original Transformer model that has not been improved by adding the improvements in this paper, and only the position encoding is changed to a learnable absolute position encoding. The experiments use a patch size of $4 \times 16 \times 16$ (dimensions are time, height, and width, respectively, and the same below) by default without any additional description, and fill it with 0 when it is insufficient, and the number of heads for multi-head attention is 4 by default. The detailed hyper-parameter settings shared in the model training are shown in Table 3, and the hyper-parameter settings in the table are used in all the following experiments without any additional description. The AdamW optimizer was used for model training. To mitigate model overfitting, the DropPath regularization trick and Weight Decay were used in this paper, and all models were trained uniformly for 30 rounds. All the experiments in this section train the models on the training set and all the metrics are calculated on the validation set.

4.2.1. Experiments on loss function for SST prediction

The experiments in this section investigate the effect of different loss functions on the effect of SST prediction. The deep learning model firstly needs to determine a suitable optimization strategy, in which the design of loss function is an important part, which directly affects the effect of optimization. In this section of the experiment, the depth of the model is set to 4, and the output length is set to 1 month, i.e. the next month prediction is performed. Based on the periodic characteristics of SST change, the input length is directly set to 2 years, i.e. 24 months, and three models are trained using the MSE loss, the MAE loss, and the loss function combining MSE and MAE proposed in this paper, respectively, and the value of μ in Equation(13) is set to 0.2, and the results are shown in Fig. 5.

From the figure, we can see that the training process is unstable when using the MSE loss function alone, the training process is more stable

when using the MAE loss function alone, but the optimization effect is poor at the late stage of the training process, and the optimal effect is achieved by using the combined MSE and MAE loss function, and the final effect of the MSE combined with the MAE loss function is significantly better than that of using the MSE or the MAE alone. The experiments in the following sections are based on the combined loss function of the MSE and MAE, which is the most effective way to optimize the training process. The experiments in the following sections are based on MSE and MAE loss functions.

4.2.2. Data input length experiment

The experiment also needs to determine the optimal historical data input length required to predict the SST for 1–24 months in the future, and the input length needs to take into account the efficiency of the experiment and the accuracy of the prediction in the later paper. The historical input length is set to 1–24 months respectively, and the output length is fixed to 24 months, and the multi-output prediction strategy is used to study the effect of different input months on the prediction effect of the model, and the results are shown in Fig. 6.

As can be seen from the figure, the two metrics, RMSE and MAE, fall faster when the input length is 1–6 months, and fluctuate down from 6 to 24 months. SSIM, on the other hand, rises faster when the input months are 1 to 6, and then fluctuates up. Finally, the overall effect is best when the input month is 24. Considering the experimental results and the periodicity of SST in the actual problem, the 24-month history input month is finally chosen for subsequent experiments.

4.2.3. Mode pre-training experiments

We conduct pre-training experiments on our SST prediction model. The experiments are divided into two groups: one group uses the simulation test outputs of the climate system model for model pre-training first, and then fine-tunes the model on real observational data. The other group is trained directly with real observational data, and the depth of the model is 4 layers in both cases, and the results on the validation set of observational data are shown in Fig. 7.

From the results, it can be seen that the pre-training and migration learning methods are able to reduce the RMSE and MAE errors of the final model prediction and improve the SSIM of the model prediction, in addition, the pre-training model training process is more stable and faster convergence, and less fluctuation at the late stage of training.

4.3. Ablation study

Ablation experiments are performed for each model improvement in the previously proposed model in order to validate the soundness of each module and to determine the parameters of the model. The optimizer and training steps for the experiments are the same as those used in the

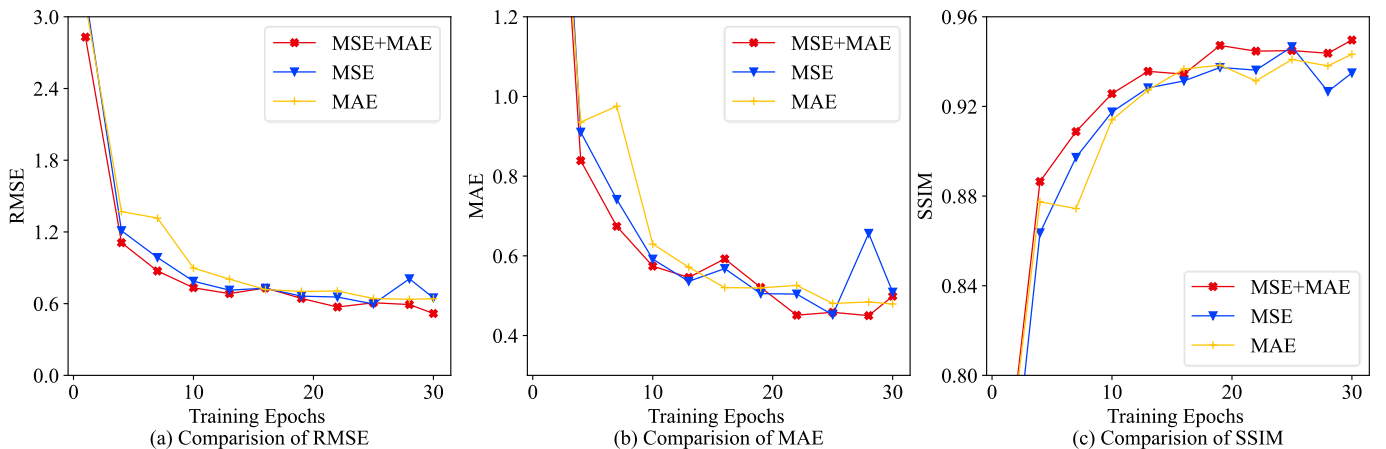


Fig. 5. Comparison of different loss functions.

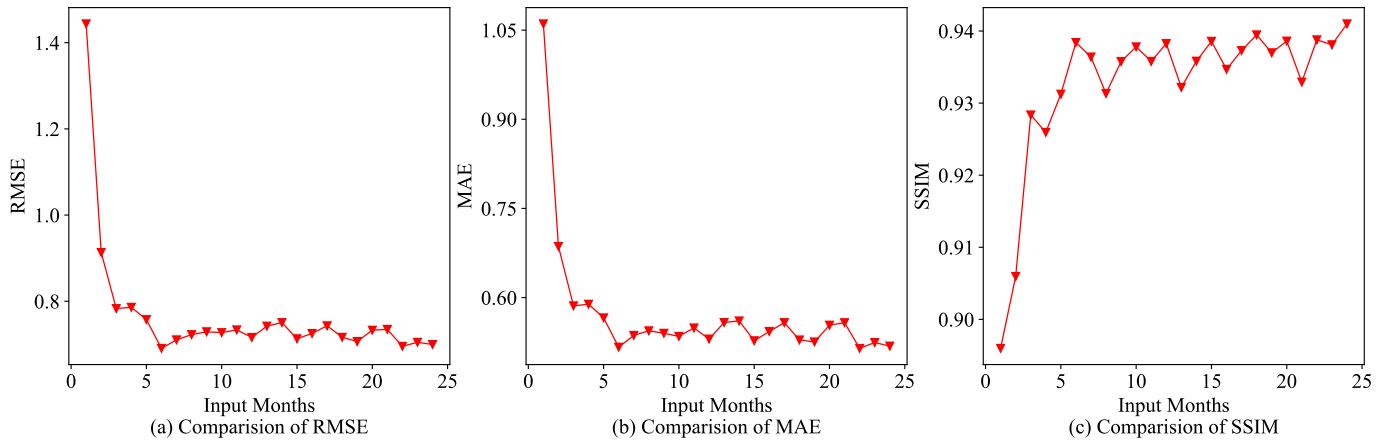


Fig. 6. Comparison of different input months.

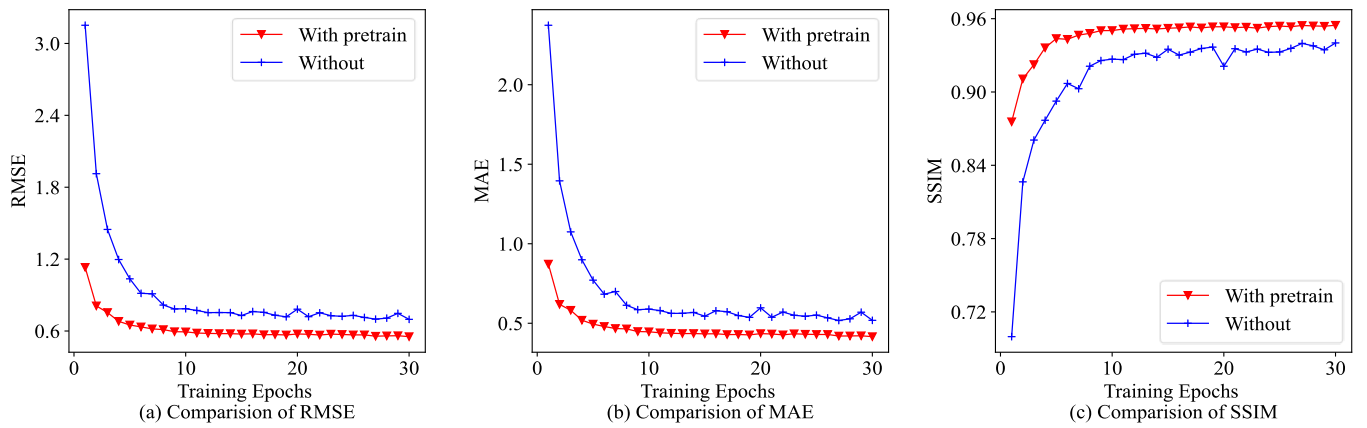


Fig. 7. Direct training vs. fine-tuning.

base experiments, and all convolution kernels are empirically set to a size of 3×3 and a number of 32 when not specified. The experiments are all performed by training the model on the training set, and the metrics are computed on the validation set.

4.3.1. Effectiveness of shallow feature extraction and fusion module

The shallow feature extraction and fusion module is based on the channel attention mechanism and is used to extract shallow features as well as to fuse SST and sea level pressure data features. This experiment is used to demonstrate the usefulness of this module. In this experiment, the history input length is fixed to 24, the model depth is set to 4, the parameter r is set to 32, and c' is set to 8. The results are shown in Fig. 8.

From the results in the figure, it can be seen that the shallow feature extraction and fusion module facilitates the rapid convergence of the model in all three metrics, and the final RMSE and MAE are lower than those of the model without the module. On the SSIM metrics, the module favors the final SSIM, which shows that mining shallow semantic information when performing SST prediction does help to improve the model prediction.

4.3.2. Experiments on effectiveness of convolutional FNN

In this section, ablation experiments are performed on Convolutional FNN layers, where we improve the FNN layer in the sublayer structure of the Transformer by inserting a convolutional layer in between the two fully-connected layers to fuse the information from neighboring patches. In the experiments, the history input length is set to 24, the model depth is 4, the convolutional kernel size is empirically set to 3×3 , the number of convolutional kernels is 32, and the d-value in the

first fully-connected layer is 12. We do not add the shallow feature extraction and fusion module in this experiment, and we only compare the effect of using the Convolutional FNN and the FNN. Additionally, we present the effects of adding convolution operations at three other positions mentioned in the previous section. The experiments results are shown in Fig. 9.

From Fig. 9, it can be seen that adding convolutions at different positions all have some influence on the optimization process of the model, but from the final results, the three methods of adding convolutions, Pre-Conv, Mid-Conv and Post-Conv, do not have any obvious improvement in effect compared to the model without convolutions, and are even weaker than the model without convolutions; however, the method of Inner-Conv is able to significantly improve the prediction of the model, reduce the RMSE and MAE, and improve the SSIM metric value, in addition, notice that inserting the convolution into the two fully-connected layers of the FNN sublayer, and at the same time shrinking the output size of the first fully-connected layer, has a smaller number of parameters compared to the other three methods as well as the original model parameters, and for this reason, the depth of the model of the method was increased to 8 layers in order to let it be approximately the same as that of the original Transformer model has approximately the same number of parameters, as shown by the red curve in Fig. 10, the model outperforms the original Transformer in all three metrics at depth 8.

On a specific example, the prediction results of a month over the previous 18 months on the validation set are randomly selected, and the heat maps are obtained by using the predicted values minus the true values as shown in Figs. 10 and 11. As can be seen from Fig. 10, if the convolution operation is not added, there are obvious demarcation lines between different patches in the prediction results, and there are

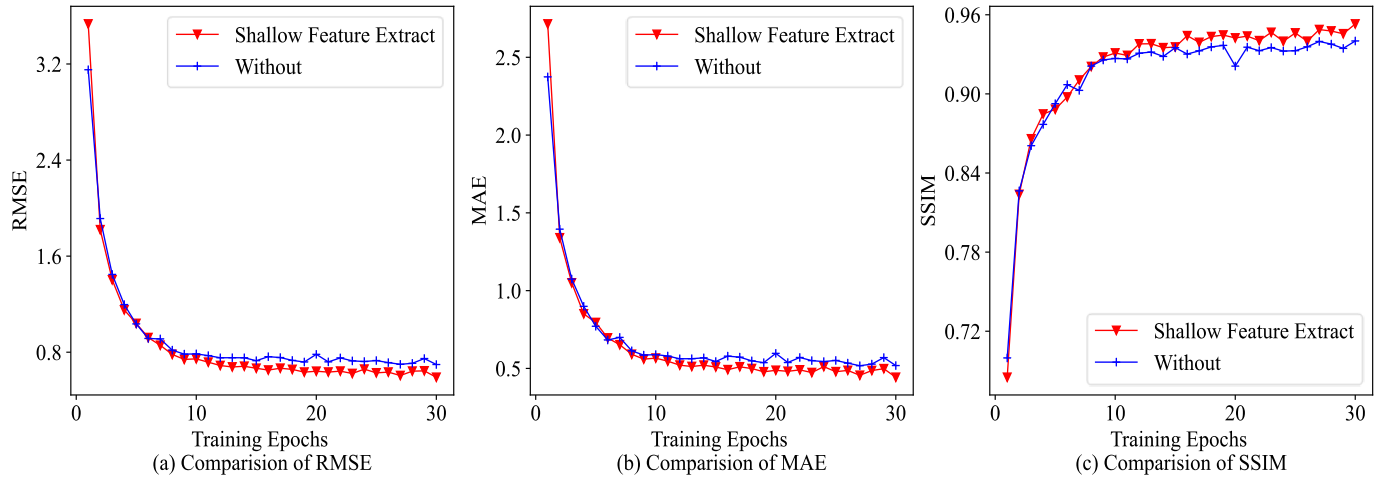


Fig. 8. Feature extraction and fusion module ablation experiment results.

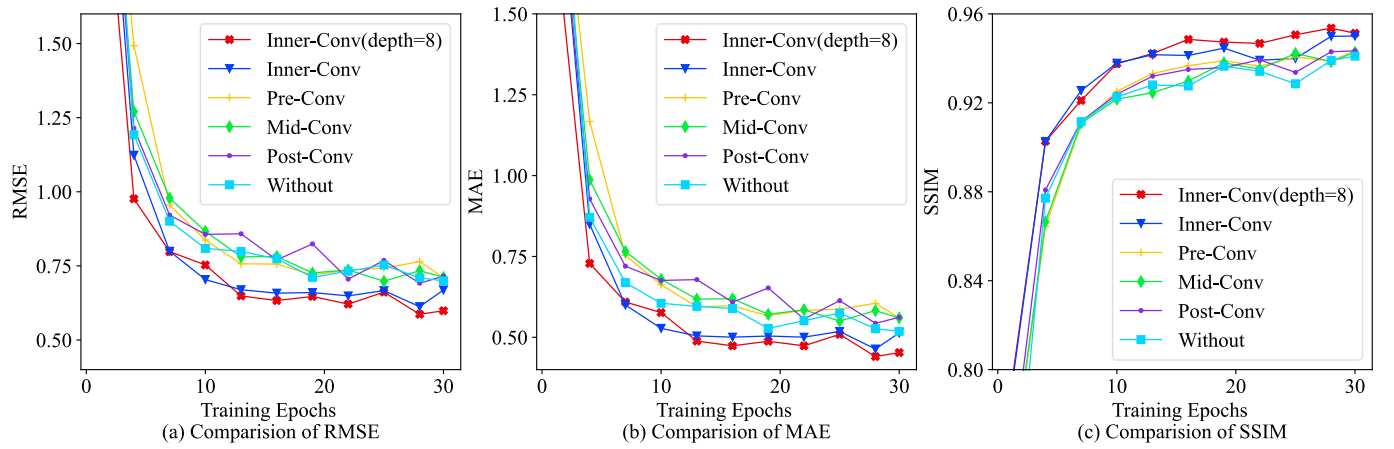


Fig. 9. Comparison of adding convolution at different locations.

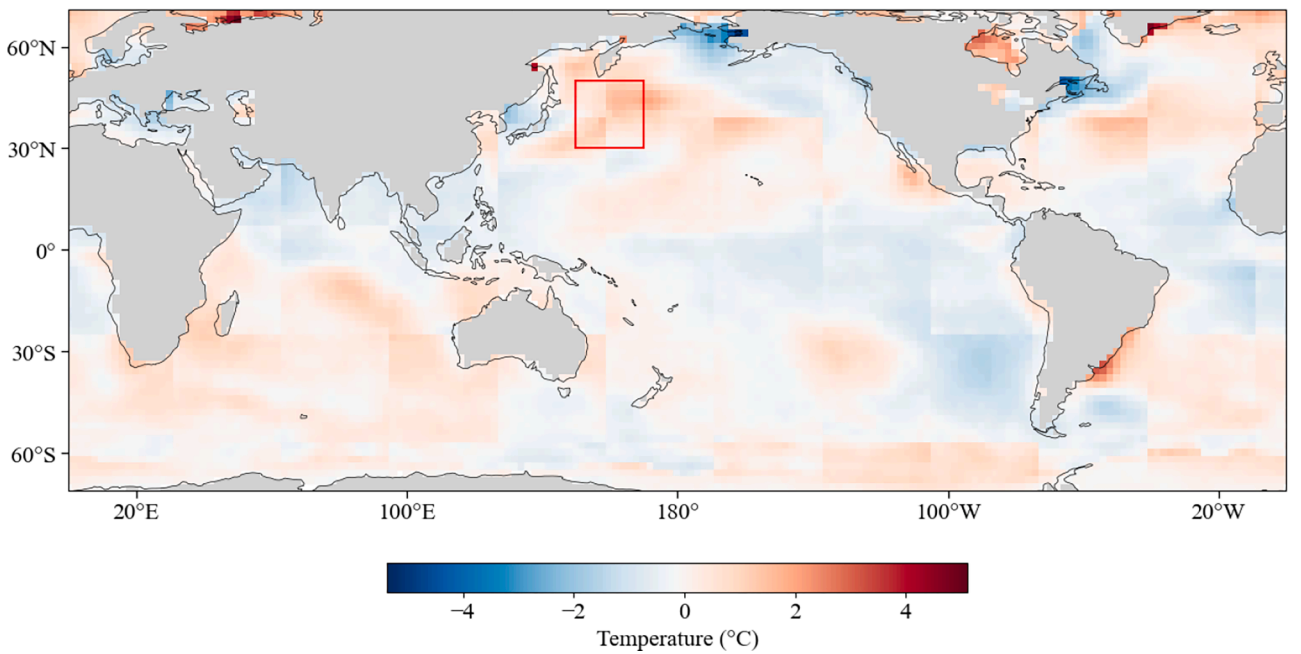


Fig. 10. Visualization of prediction error without adding convolution.

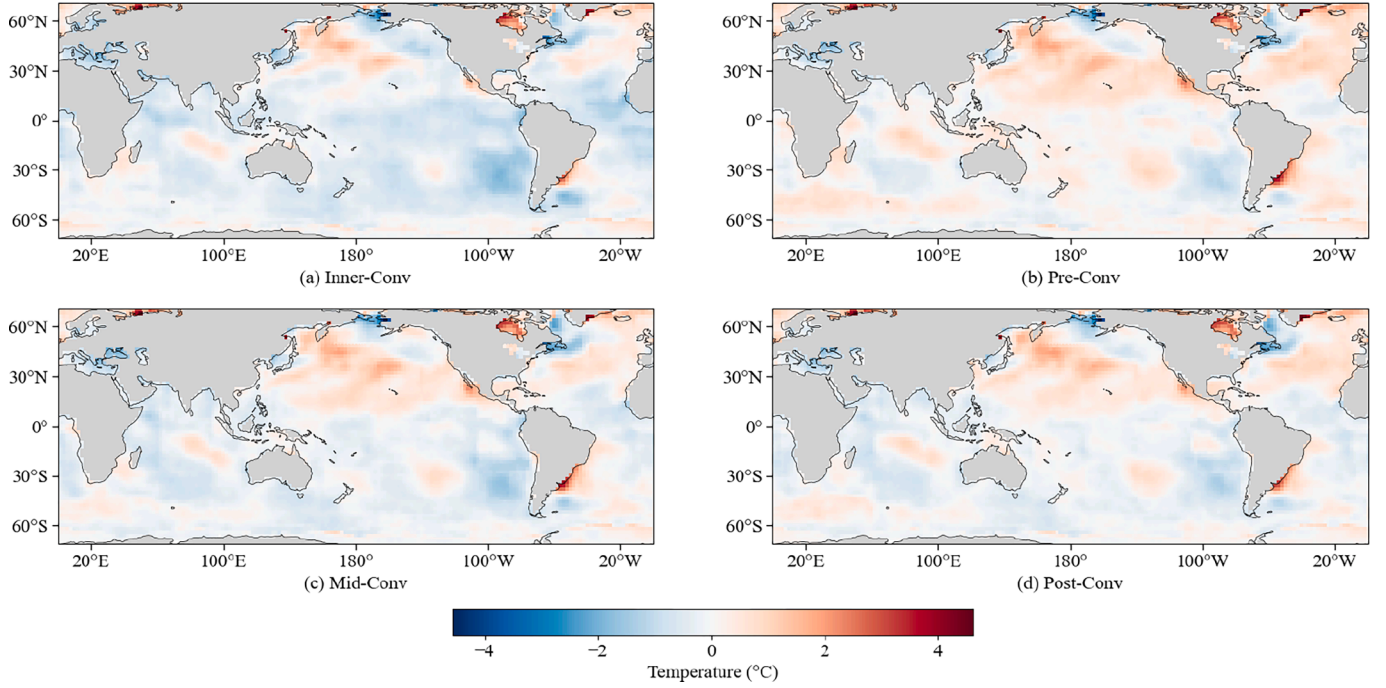


Fig. 11. Visualization of prediction error after adding convolution at different locations.

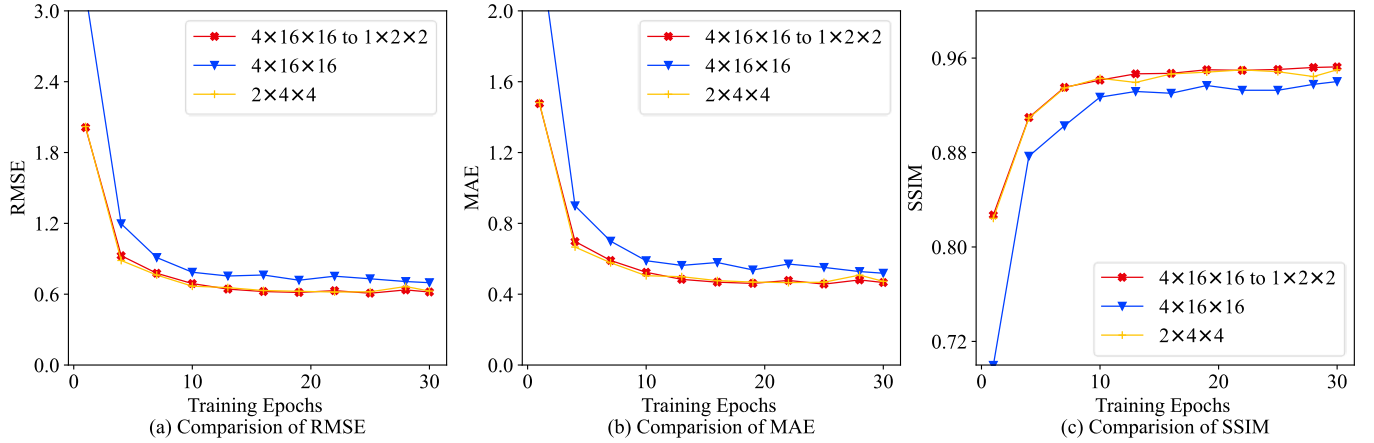


Fig. 12. Comparison of different patch sizes.

obvious faults between the predicted textures of certain demarcation lines, as shown in the red box region of the figure. While the error visualization results after adding the convolution are shown in Fig. 11, the four plots correspond to the four different additions in the previous description, and it can be seen that the different additions are able to alleviate the demarcation line between patches, and the texture of prediction errors between different patches can basically match, which indicates that the convolution layer is beneficial to the extraction and fusion of local features. The worst effect is the Pre-Conv way of adding the convolutional operation before the attention layer, and the optimal is to add the convolutional operation between two fully connected layers, which is exactly the Convolutional FNN layer.

4.3.3. Experiments on image patch size

In this subsection, experiments are conducted on the size of image patches, and the model is uniformly set to 4 layers without adding the Feature Extraction and Fusion module and Convolutional FNN layer. The experiments are divided into two groups, one group is fixed patch size for each layer, which is $4 \times 16 \times 16$ and $2 \times 4 \times 4$, and the size of the hidden layer is 128 and 96, respectively; one group gradually re-

duces the patch size, and the patch size change process is $4 \times 16 \times 16$, $3 \times 8 \times 8$, $2 \times 4 \times 4$, and $1 \times 2 \times 2$, and the first fully-connected layer of the Convolutional FNN with d-value is sequentially is 12, 6, 2, 2, and the size of the hidden layer is 128, 128, 96, 64 in order, in which the first layer uses global attention, the second to the fourth layer uses local attention, and the attention window is uniformly set to $6 \times 5 \times 10$, and this window size is equivalent to the size of the global attention window of the first layer, and the experimental results are shown in Fig. 12.

From the figure, it can be seen that using a smaller patch is favorable for the fast convergence of the model and the improvement of the final SSIM metrics, but the use of a small patch leads to an increase in the computational attention complexity. The method of gradually shrinking the patches is close to the effect of using smaller patches.

4.3.4. The number of spatiotemporal transformer encoder layers

Through the above three experiments, it can be seen that all three improvements have gained the prediction effect on the original Transformer model, and in this section, all three improvements are added to the model to investigate the effect of these three improvements on the model with different depths, while the original Transformer is used as

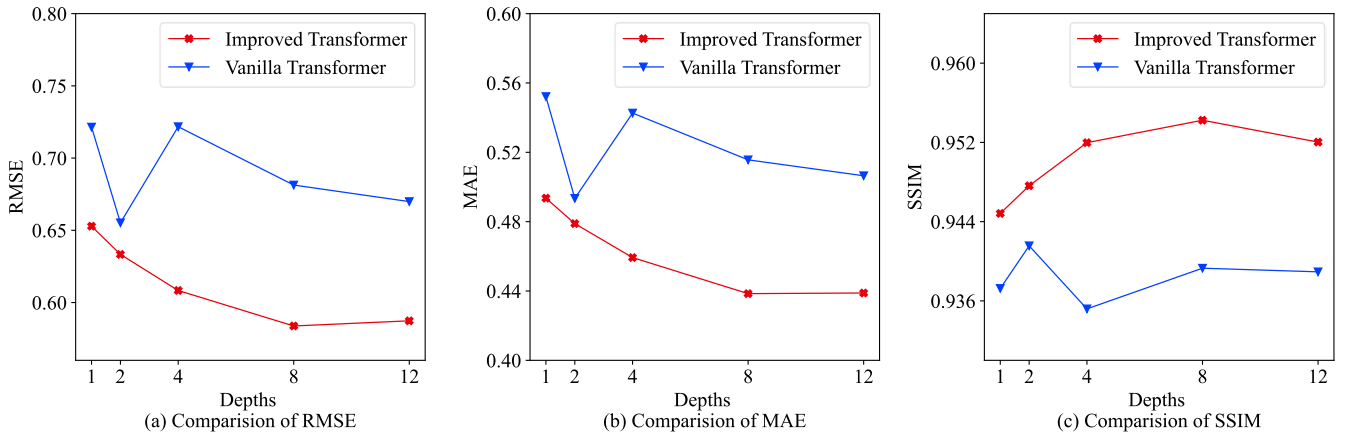


Fig. 13. Comparison of different depths.

Table 4
Hyperparameter setting of CNN-AT transformer model.

Parameters	Value
Number of stages in the spatiotemporal Transformer encoder	4
Number of stacked MHA + Convolutional FFN blocks at each stage (N)	2
Number of heads in Multi-head Attention	4
Feature Extraction Fusion Module r	32
Feature Extraction Fusion Module c'	8
Number of channels in 4 stages	128, 128, 96, 64
Patch sizes in 4 stages	$4 \times 16 \times 16$, $3 \times 8 \times 8$, $2 \times 4 \times 4$, $1 \times 2 \times 2$
Convolutional FNN first fully connected layer d-value	12, 6, 2, 2
Number of Convolutional FNN Convolutional Kernels	32
Convolution kernel size	3×3
Optimizer	Adam
Epochs	30
learning rate	0.001
Weight Decay	0.05
β_1	0.9
β_2	0.999
DropPath rate	0.1

a comparison. The number of layers of the spatiotemporal Transformer encoder is set to be 1, 2, 4, 8, and 12, respectively, and the patch size change process is $4 \times 16 \times 16$, $3 \times 8 \times 8$, $2 \times 4 \times 4$, and $1 \times 2 \times 2$, with the same d-value as the experiments in the previous section, and the size of the hidden layers is 128, 128, 96, and 64 in order, and the patch is shrunk every two layers (i.e., one stage) when there are less than 8 layers. The experimental results are shown in Fig. 13.

From the figure, it can be seen that the CNN-AT Transformer outperforms the original Transformer at all depths, and the prediction error of the original Transformer does not necessarily decrease with deeper depths, while the prediction error of the CNN-AT Transformer gradually decreases, and overall, the CNN-AT Transformer's RMSE and MAE errors can be reduced by about 10 %, which indicates that the method of adding convolutional layers and shrinking patches is beneficial to Transformer's error reduction and improving the training stability of the model. In addition, when the model depth is increased to 12 layers, the prediction error of CNN-AT Transformer slightly improves, which indicates that 8 layers are a better choice.

The above ablation experiments results show that the shallow feature extraction and fusion module is conducive to improving the model convergence speed and prediction accuracy, the insertion of convolutional layers in the Transformer is conducive to the extraction and fusion of local features and the location of the inclusion has a greater impact on the results, the model structure of gradually reducing the size of the patches and the local window auto-attention are conducive to the model's learning of local features, and the CNN-AT Transformer outperforms the original Transformer model in different number of layers. Based on the results of the ablation experiments, we set the number of

stages of the CNN-AT Transformer model, which incorporates the shallow feature extraction and fusion module and the Convolutional FNN, to 4. The depth of each stage is set to 2, and the initial patch size is set to $4 \times 16 \times 16$, which is gradually reduced at each stage. The detailed hyperparameter settings are shown in Table 4.

4.4. Performance comparison

To further verify the effectiveness of the proposed CNN-AT Transformer model, we conducted a series of experiments and carried out a comparative analysis with other representative baseline models for SST prediction. These baseline models mainly include traditional CNN, LSTM, and Transformer models, as well as ConvLSTM (Hao et al., 2023), PredRNN (Qiao et al., 2023), MIM (Wang et al., 2019), TempProNet (Chen et al., 2024), and TransDtSt-Part (Dai et al., 2024) models. The following will present a comparative analysis from the aspects of medium- and long-term sea surface temperature prediction and short-term sea surface temperature prediction respectively.

4.4.1. Medium- and long-term SST prediction

On the aforementioned datasets, we evaluated the CNN-AT Transformer model alongside several baseline models. All models were trained using the weight parameters that achieved the highest validation performance. The following will present the experimental results on the test sets. We first use the 24-month monthly mean SSTs as input and employ several models to predict the monthly mean SST for the next 6, 12, and 24 months, respectively. The scores of each model are then

Table 5
Comparison of monthly mean SST predictions.

Model	RMSE	MAE	SSIM
CNN	0.6376	0.4702	0.9516
LSTM	0.9268	0.6832	0.8999
ConvLSTM	0.8701	0.6657	0.9376
PredRNN	0.7400	0.5533	0.9440
MIM	0.8461	0.6431	0.9400
Transformer	0.6752	0.5153	0.9406
TemproNet	0.6263	0.4587	0.9533
TransDtSt-Part	0.6317	0.4721	0.9476
CNN-AT Transformer	0.6084	0.4457	0.9558

calculated on three metrics, and their average values are computed to obtain the overall metric scores. The results are shown in Table 5.

From the Table 5, it can be seen that although the CNN model does not consider the temporal relationship of the SST data, it is able to achieve better results due to the CNN's strong ability to extract spatial information and it is easy to be trained, which shows that spatial dependence is very important in the prediction of SSTs. Whereas, the absence of spatial structural information in the LSTM has a great negative impact on the prediction results, and the three metrics are all significantly weaker than the other models. It can be seen from this that the spatial relationship has a greater impact on the SST prediction than the temporal relationship. ConvLSTM, which is a combination of CNN and LSTM, is able to achieve a better performance than LSTM, precisely because of the spatial features it learns, but its performance is not as good as that of the CNN model. Although PredRNN and MIM can capture more spatial and temporal details than ConvLSTM, they can achieve a better performance than ConvLSTM. ConvLSTM to achieve better results, but overall they are also inferior to CNN models, which also illustrates the shortcomings of RNN-like models.

The Transformer model and the Transformer-based models, TransDtSt-Par, TemproNet, and CNN-AT Transformer, have achieved relatively high prediction precision due to their good capabilities in extracting long-term trends. Among them, the TransDtSt-Par model adopts a spatiotemporal attention distillation mechanism on the basis of the Transformer, enabling long-sequence modeling. Therefore, its prediction accuracy is better than that of the traditional Transformer model. However, its generative decoding method is prone to error accumulation. The TemproNet model employs an encoder-decoder structure. It integrates the Transformer with the CNN network to achieve seawater profile prediction and has obtained better prediction results than the Transformer. Nevertheless, it does not design a constraint mechanism specifically for the spatial continuity of SST field. Our model CNN-AT Transformer uses a CNN-based joint attention mechanism to mine and fuse the shallow semantic features of different oceanic and atmospheric features. Moreover, convolutional layers are embedded in the Transformer sub-layers to mine and fuse local feature information. This enables it to capture the cross-ocean long-distance correlation characteristics unique to events such as the El Niño - Southern Oscillation (ENSO).

As a result, it has high prediction accuracy, which is also the reason why our model CNN-AT Transformer outperforms other baseline models in three evaluation metrics.

Table 6 shows the scores of several models when predicting SST at prediction horizons of 6 months, 12 months, and 24 months. As can be seen from the table, as the prediction horizon increases, the scores of the several models on the MAE and RMSE indicators increase, while the scores on the SSIM indicator decrease, indicating that the prediction accuracy decreases, which is consistent with expectations. Our proposed CNN-AT Transformer model performs optimally on all metrics, proving the effectiveness of our proposed model.

Fig. 14 shows the results of different models for RMSE, MAE, and SSIM on different predict step size, respectively. The results show that for RMSE and MAE, the prediction errors of all models are not monotonic. This is due to the fact that the SST variations have a clear yearly and seasonal periodicity, which the models learn and utilize for their predictions. For the SSIM metric, it can be seen that when the prediction step size is 1 month, 2 months, and 3 months respectively, the SSIM values are relatively high, and then gradually decrease. However, the SSIM value does not decrease monotonically, but fluctuates slightly over time.

As can also be seen in Fig. 14, when the prediction step size is 1 to 3 months, the prediction accuracy of the CNN model is obviously better than that of RNN-like models, and it shows certain fluctuations with increasing prediction step size. This indicates that the CNN model can also learn long-term dependencies. For the RMSE and MAE metrics, the prediction results of our model at different prediction step sizes are better than those of RNN models, but the performance is not as good as CNN in some prediction step sizes. In terms of the SSIM metric, our model is better than other models.

From the figure, it can be seen that the seasonal variation of the SST has significant periodicity. The experimental results show that RMSE and MAE do not increase monotonically with the leading months, but show a fluctuating trend, which is related to the effective learning of seasonal cycles by the model. The prediction errors for summer and winter in the northern hemisphere may be reduced due to the strong regularity of the seasonal signals. The model fully learns the superposition effect of seasonal cycles by inputting 24 months of historical data. In both the 6-month and 12-month predictions, the SSIM values were higher than 0.95, indicating that the model accurately reproduced the seasonal driven temperature distribution pattern while maintaining spatial structure.

4.4.2. Medium- and long-term prediction of El Niño index

Since the magnitude of the El Niño index directly reflects the degree of SST anomalies, we predicted the medium- and long-term El Niño index based on the previous SST prediction results. We also calculated the correlation between the predicted El Niño index and the actual El Niño index. The larger the correlation, the more accurate the predicted value is, which represents the prediction accuracy to some extent. Table 7 shows the prediction correlations of several models with prediction step lengths of 6, 12, and 24 months. It can be seen from the table

Table 6
Comparison of SST predictions results with different prediction steps.

Prediction Step	6 months			12 months			24 months		
Metrics	RMSE	MAE	SSIM	RMSE	MAE	SSIM	RMSE	MAE	SSIM
CNN	0.6315	0.4648	0.9517	0.6253	0.4553	0.9522	0.6444	0.4713	0.9509
LSTM	0.9423	0.6988	0.8907	0.9144	0.6701	0.9006	0.9085	0.6635	0.9059
ConvLSTM	0.8491	0.6473	0.9378	0.8622	0.6584	0.9380	0.8934	0.6850	0.9377
PredRNN	0.6940	0.5074	0.9460	0.7853	0.6003	0.9416	0.7890	0.6027	0.9416
MIM	0.8490	0.6397	0.9413	0.8198	0.6337	0.9417	0.8197	0.6350	0.9421
Transformer	0.6714	0.5011	0.9479	0.7135	0.5413	0.9446	0.7612	0.5776	0.9418
TemproNet	0.6292	0.4380	0.9541	0.6423	0.4681	0.9525	0.7063	0.5186	0.9443
TransDtSt-Part	0.6402	0.4451	0.9508	0.6610	0.4748	0.9455	0.7115	0.5218	0.9432
CNN-AT Transformer	0.5995	0.4366	0.9556	0.6039	0.4410	0.9562	0.6225	0.4561	0.9546

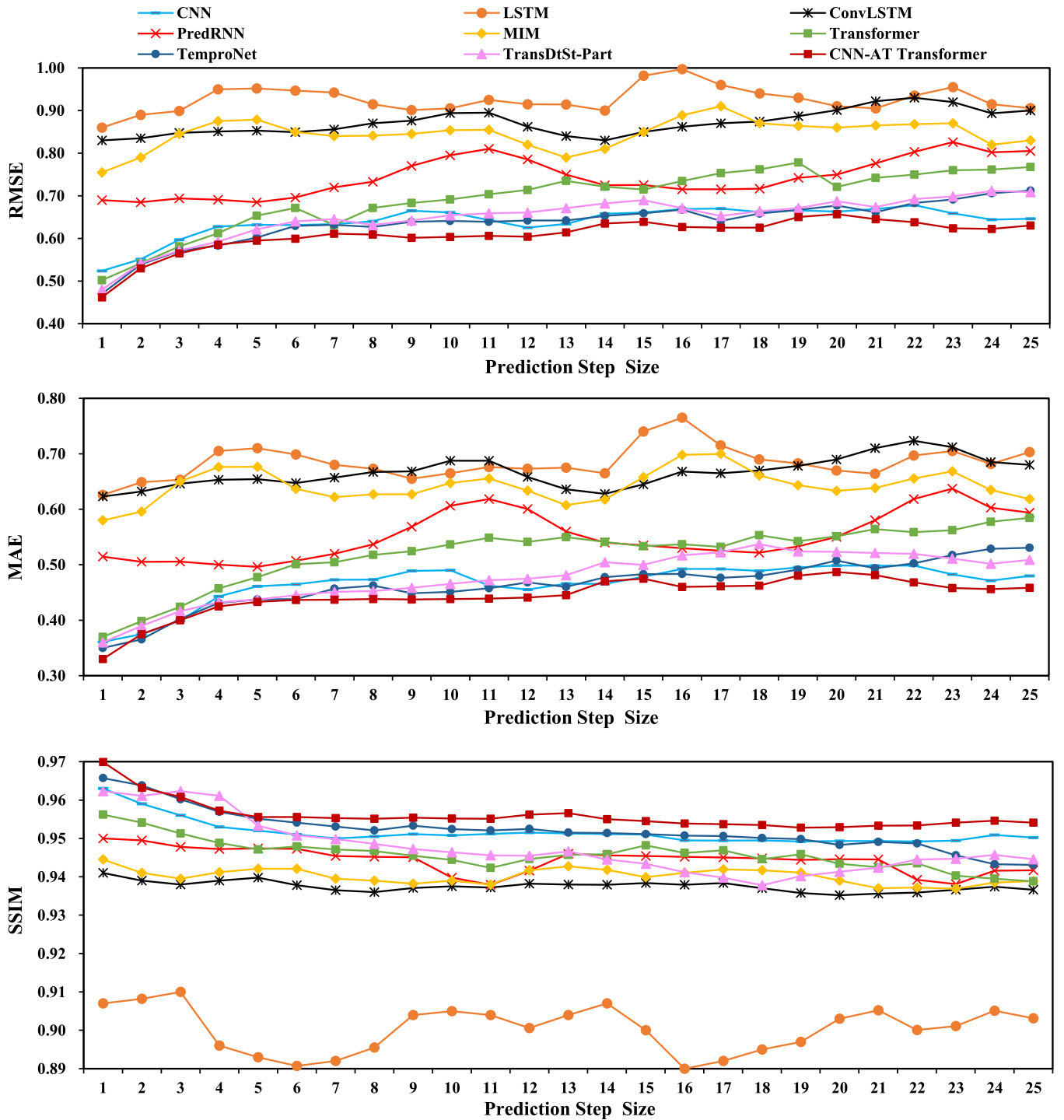


Fig. 14. The scores of several models on three metrics vary with the prediction step size.

that as the prediction step length increases, the prediction accuracy decreases significantly. When the step length is 6 months, several models have a relatively strong correlation. When the step length is 12 months, there is a certain correlation, while when the prediction step length is 24 months, the correlation is relatively weak. At the same time, it can be seen that the prediction accuracy of each model is similar to the performance of the medium- and long-term SST prediction. The CNN and Transformer-based models perform better than the RNN-based models. Among the RNN-based models, the PredRNN model is the best and the LSTM model the worst. Our CNN-AT Transformer model is still the best.

Fig. 15 shows the variation of the Nino3.4 index from 1980 to 2020 predicted by our CNN-AT Transformer model on the test set for prediction steps of 12 and 24 months. As can be seen from the figure, when the prediction step is 12 months, the variation trends of the predicted values and the actual values of the Nino3.4 index are consistent, indicating that the model can basically predict the variation of the Nino3.4 index. However, its prediction ability for the peak values is insufficient. When the prediction step is 24 months, the prediction effect is relatively poor, suggesting that the prediction ability of the model still needs to be improved.

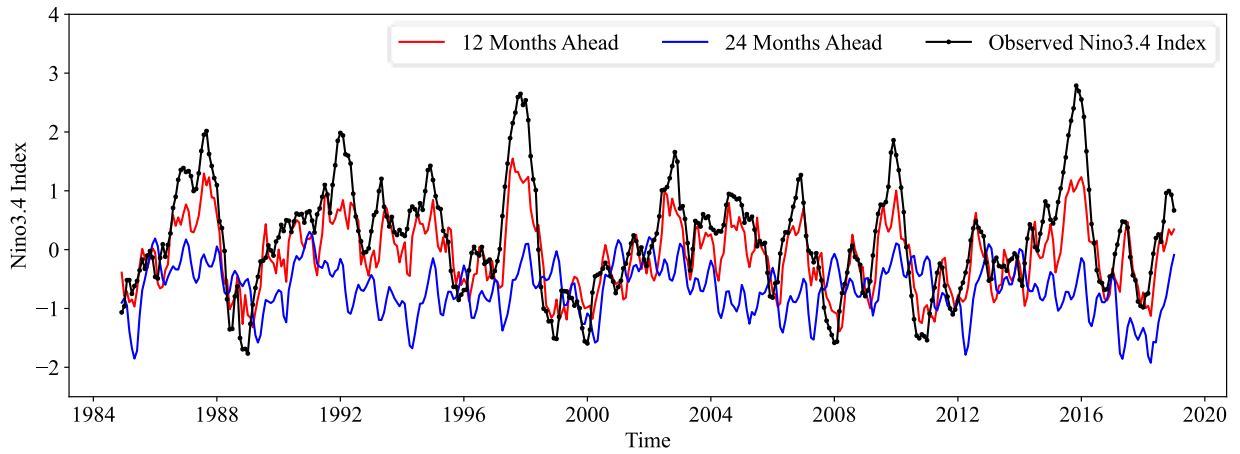


Fig. 15. Comparison of different depths.

Table 7
Comparison of El Niño 3.4 index correlation coefficient.

Ahead of schedule	6	12	24
CNN	0.7243	0.6022	0.2426
LSTM	0.4455	0.3862	0.2447
ConvLSTM	0.6432	0.4215	0.1951
PredRNN	0.7082	0.4826	0.1967
MIM	0.6724	0.4680	0.1481
Transformer	0.6934	0.4935	0.2126
TemproNet	0.7337	0.6147	0.2445
TransDtSt-Part	0.7485	0.6377	0.2575
CNN-AT Transformer	0.7674	0.6570	0.2609

Table 8
Comparison of daily mean SST prediction.

Model	RMSE	MAE	SSIM
CNN	0.7471	0.5848	0.9050
LSTM	1.1995	0.8460	0.8662
ConvLSTM	0.9745	0.7420	0.8980
PredRNN	0.7164	0.5183	0.9220
MIM	0.9265	0.7087	0.9089
Transformer	0.6915	0.5418	0.9107
TemproNet	0.6778	0.4845	0.9216
TransDtSt-Part	0.7512	0.5963	0.9242
CNN-AT Transformer	0.6500	0.4769	0.9271

4.4.3. Short-term SST prediction

In order to comprehensively evaluate the prediction capability of the models, we used the daily mean SSTs of the past 30 days as the input to predict the daily mean SSTs for the next 1 to 30 days. Since the data used in the experiment is relatively large, the models were not pre-trained. The experimental results of several models on the three evaluation metrics of RMSE, MAE and SSIM are shown in Table 8.

From the results in Table 8, it can be seen that due to the lack of spatial information, the LSTM model performs significantly worse than the other models that have the ability to extract spatial features. The CNN model is slightly inferior to the PredRNN model only in terms of RMSE and MAE, but it performs worse than both PredRNN and the MIM Model in terms of SSIM. However, it outperforms the ConvLSTM model on all three metrics, indicating that CNN can also learn long-term dependencies well in short-term SST prediction. The ConvLSTM model has better prediction ability than the LSTM model because it integrates convolution operations. The PredRNN model improves upon the deficiencies of ConvLSTM, and its results are significantly better than those of the ConvLSTM model. The overall performance of the MIM model is between that of PredRNN and ConvLSTM. Both the TransDtSt-Part

Table 9
Ocean regions selected for short-term SST prediction.

Number	Sea region	Latitude and longitude range
1	Western Pacific Ocean	0°-16°N; 125°E-165°E
2	Eastern Equatorial Pacific Ocean	5°S-5°N; 140°E-100°W
3	Pacific Ocean	5°S-5°N; 50°E-70°E
4	North Atlantic Ocean	1°S-51°N;]] 95°W- 180°W

and TemproNet models are improved versions based on the traditional Transformer model. Therefore, their prediction accuracies are superior to those of the Transformer model and RNN-based models. The prediction accuracy of the TemproNet model is slightly higher than that of the TransDtSt-Part model. However, in the short-term SST prediction, the attention distillation strategy of TransDtSt-Part overcompresses the time information, resulting in the loss of key transient signals. Moreover, partial stacking connections may introduce the problem of gradient fragmentation, thereby affecting the stability of training. By hierarchically combining the Transformer with a CNN decoder, TemproNet reduces the prediction error, which proves the effectiveness of the hybrid architecture in vertical profile prediction. Our CNN-AT Transformer model significantly outperforms other models in three metrics, indicating that our model also has high prediction accuracy for short-term SST. The main reason is that the CNN-AT Transformer model effectively balances local details and global dependencies through shallow feature extraction and a segmented attention mechanism. It can capture small-scale variations in SST, such as fronts and vortices, thus achieving better prediction accuracy.

Fig. 16 shows the scores of several models on three indicators, RMSE, MAE and SSIM, when the prediction step size increases. As can be seen from the figure, with the increase of prediction step size, the RMSE and MAE values of several models increase slowly, while the SSIM decreases slowly, which indicates that the prediction difficulty increases as the prediction step size increases, which is consistent with expectations. Our model, CNN-AT Transformer, is superior to other models at all prediction step sizes, which also proves the effectiveness of our model.

We selected several typical ocean regions shown in Table 9 for prediction experiments to further verify the effectiveness of our model. Table 10 shows the scores of several models on the RMSE and MSE metrics. It can be seen that the prediction accuracy of the CNN-AT Transformer model in the three ocean areas is better than other baseline models, and slightly lower than the PredRNN model in the Western Pacific region, which further proves the effectiveness of our model.

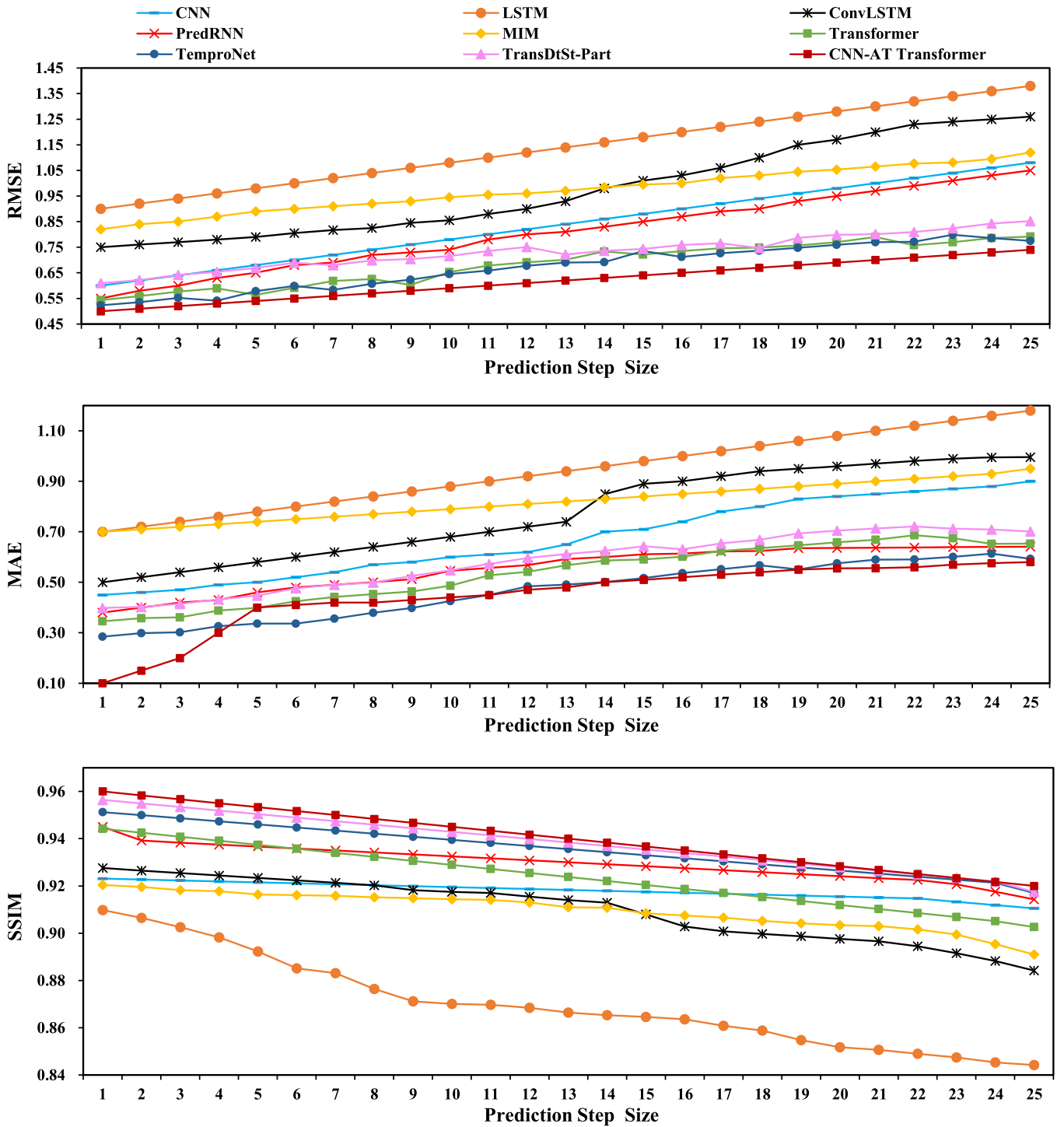


Fig. 16. The scores of several models on three indicators with prediction step size.

From Table 10, it can be seen that the RMSE of the North Atlantic is slightly higher than that of other regions, possibly due to the influence of mid latitude dynamic processes such as the North Atlantic Oscillation. The SST of the upwelling region is affected by local wind fields and ocean dynamic processes, with drastic changes and large spatial gradients. The model extracts local details through shallow CNN and Convolutional FNN modules, and embeds convolutional layers in the Transformer to alleviate the boundary effect of traditional Transformers. The low error in the central equatorial Pacific region in the table verifies the adaptability of the model to such complex dynamics. The strategy of

reducing patches in stages simultaneously allows the model to capture the overall trend of the large-scale upwelling area in the early stages, and focus on local resolution in the later stages, thus balancing the modeling of global teleconnection and local processes. ENSO events involve SST anomalies and atmospheric teleconnection between the eastern and western Pacific oceans. The CNN-AT Transformer captures long-distance spatiotemporal correlations on a global scale through its self attention mechanism. The RMSE and MAE in the eastern equatorial Pacific region are the lowest, indicating that the model can effectively predict ENSO related temperature anomalies.

Table 10
Results of several models on the RMSE and MSE metrics.

Sea Region	Western Pacific Ocean		Eastern Equatorial Pacific Ocean		Pacific Ocean		North Atlantic	
Metrics	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
CNN	0.6435	0.5188	0.7415	0.5904	0.7213	0.5725	0.7709	0.6026
LSTM	0.7437	0.5938	0.7730	0.6127	1.0061	0.8128	1.3244	0.9646
ConvLSTM	0.5892	0.4669	0.7083	0.5440	0.8316	0.6374	1.0300	0.7722
PredRNN	0.5778	0.4564	0.6069	0.4738	0.7090	0.5620	0.7872	0.5983
MIM	0.8935	0.7527	0.9248	0.7614	1.0643	0.9012	1.0355	0.8265
Transformer	0.6374	0.4935	0.6484	0.5121	0.7235	0.5834	0.7556	0.6013
TemproNet	0.6135	0.4725	0.6352	0.4853	0.7446	0.5975	0.7613	0.6224
TransDiSt-Part	0.6825	0.5476	0.6935	0.5613	0.7753	0.6024	0.7975	0.6253
CNN-AT Transformer	0.5815	0.4605	0.5847	0.4530	0.7105	0.5541	0.7486	0.5737

Table 11
Model complexity analysis.

Indicator	CNN	LSTM	ConvLSTM	PredRNN	MIM	Transformer	TemproNet	TransDiSt-Part	CNN-AT Transformer
Number of parameters	1.02M	0.95M	1.05M	2.15M	8.21M	1.85M	1.79M	1.43M	1.32M
Execution time	3.36s	1.52s	32.57s	32.20s	23.93s	10.56s	9.22s	9.54s	8.99s

As shown in Table 10, we observe that the PredRNN model performs slightly better than the CNN-AT Transformer model in terms of RMSE and MAE metrics in the Western Pacific and some other Pacific regions. This difference is chiefly attributable to regional dynamics and the distinct architectural adaptability of the two models. PredRNN, based on an RNN structure, captures short-term local dynamics through ConvLSTM's gating temporal modeling, demonstrating stronger adaptability in the Pacific and Western Pacific regions, which are strongly influenced by monsoons and ocean currents. By contrast, the CNN-AT Transformer is tailored for long-range teleconnection feature extraction. Its strategy of gradually reducing patch size and the global-local attention switching mechanism perform more robustly in cross-regional, multi-time-scale prediction tasks, giving it an advantage in regions involving long-distance teleconnections. However, it slightly weakens its ability to capture details in local high-frequency short-period predictions. Additionally, the denser data coverage in the Western Pacific helps amplify PredRNN's advantages. Nevertheless, the overall performance gap remains narrow. Considering the overall accuracy, stability, complexity, and adaptability of the models, the CNN-AT Transformer retains several overarching advantages.

4.4.4. Model complexity analysis

To further evaluate the performance of the models, we conducted a comparative analysis of their complexity. We calculated the total number of parameters for each model by computing the number of parameters in each layer. Each model was run 100 times for prediction. In each run, a sample spanning 24 months was input, and the sea surface temperatures for the next 24 months were output. Based on this, we obtained the execution time of each model. Table 11 shows a comparison of the number of parameters and execution time of several models.

From Table 11, it can be observed that the total number of parameters of our CNN-AT Transformer model is slightly more than those of the CNN, LSTM, and ConvLSTM models, but less than those of other baseline models. This indicates that our model has a relatively low complexity. In terms of running speed, the CNN-AT Transformer model is significantly faster than the ConvLSTM, PredRNN, and MIM models, while being slightly slower than the CNN and LSTM models. This is because our model has the ability to execute in parallel, and its strategy of gradually reducing the patch size reduces the computational complexity of the model. This also shows that the CNN-AT Transformer model can achieve the best prediction accuracy with relatively low complexity and fast running speed, striking a good balance among complexity, running speed, and prediction accuracy. This characteristic gives it practical deployment potential in resource-constrained systems. For example, it can be deployed on edge devices such as ship-borne meteorological stations and

buoy sensors, thereby avoiding problems such as insufficient real-time performance and memory overflow caused by an overly large model.

To expand deployment capabilities in extreme resource environments, there are two lightweight improvement schemes: Progressive Patch Pruning, which maintains the original partitioning strategy during training and fixes the final partitioning size to $2 \times 4 \times 4$ during deployment, reducing the number of attention heads by 50 %; Dynamic Sparse Attention introduces an attention mask based on regional temperature gradients in the Transformer encoder to enable global attention for equatorial convective regions and local window attention for high latitude stable regions.

4.5. Discussion

4.5.1. Uncertainty estimation of the model

To further validate the reliability and trustworthiness of the proposed model in practical applications, Monte Carlo Dropout (MC Dropout) was introduced during the inference phase of the CNN-AT Transformer to quantify predictive uncertainty. Specifically, the dropout layers were kept active during testing, enabling stochastic neuron deactivation. For each input sample, 20 independent forward passes were performed, and the mean of these predictions was taken as the final output, while the standard deviation served as an estimate of prediction uncertainty.

Experimental results show that, under conventional inference (i.e., without MC Dropout), the model produces deterministic predictions with generally high accuracy, typically selecting the best single prediction as the final output. In contrast, with MC Dropout enabled, the model yields slightly higher average RMSE and MAE-about 0.85 % increase-due to the use of the mean over multiple stochastic predictions. This slight increase is within an acceptable range and does not indicate a significant performance drop. In fact, in some prediction time steps, the MC Dropout-enhanced model achieved even better results, demonstrating that the introduction of stochastic inference does not compromise overall model performance.

Furthermore, statistical analysis across multiple validation batches shows that the average predictive standard deviation-representing model uncertainty-remains consistently between 0.14 and 0.15. This indicates that our model exhibits low variance and high confidence in its predictions, along with strong generalization ability and stability. Overall, incorporating MC Dropout not only provides accurate prediction results but also enables intuitive and quantitative uncertainty estimation, significantly enhancing the model's practical value and interpretability in ocean forecasting and other high-risk application scenarios.

Table 12

t-test results of CNN-AT transformer compared to CNN and TemproNet models.

Metrics	CNN		TemproNet	
	RMSE	MAE	RMSE	MAE
t	3.149	3.22	4.26	3.567
p	0.0136	0.0122	0.0028	0.0073

4.5.2. Statistical significance testing of the mode

To thoroughly validate the effectiveness of the proposed CNN-AT Transformer model for SST prediction, we selected two representative baseline models, CNN and TemproNet, which have shown good performance in prediction tasks, to conduct a t-test. The CNN model has excellent spatial feature extraction capabilities and has demonstrated high prediction accuracy in medium- and long-term SST prediction. The TemproNet model integrates the advantageous structures of Transformer and CNN, and outperforms other baseline models in both short-term and medium- to long-term SST prediction, representing the current mainstream model's accuracy in SST prediction. After evaluating the performance of the test samples, we paired the CNN-AT Transformer model with the CNN and TemproNet models respectively. Based on the short-term and medium to long-term SST prediction results, a total of 20 samples were selected, and a t-test was conducted on two key indicators, RMSE and MAE, to verify the statistical differences between the models. Table 12 shows the t-test results of our model compared with CNN and TemproNet respectively.

It can be seen from Table 12 that for the CNN-AT Transformer compared with the CNN model, the t-value is 3.149 and the p-value is 0.0136 for RMSE, and the t-value is 3.220 and the p-value is 0.0122 for MAE. It passes at the 5 % significance level, indicating that there is a statistically significant difference in error control between the CNN-AT Transformer model and the CNN model. In the comparison with the TemproNet model, the t-value of the CNN-AT Transformer is 4.260 and the p-value is 0.0028 for RMSE, and the t-value is 3.567 and the p-value is 0.0073 for MAE, both reaching the 1 % significance level. These experimental results and statistical analyses both indicate that the CNN-AT Transformer significantly outperforms the representative baseline models in terms of the RMSE and MAE metrics. This conclusion is highly significant and reliable, providing a basis for the application and decision-making of the model in SST prediction.

To further validate the predictive performance of our CNN-AT Transformer model, we introduced the effect size metric, Cohen's d, to evaluate the difference of CNN-AT Transformer model against several baseline models in terms of the RMSE metric. The Cohen's d metric can effectively measure the real strength of the difference in prediction errors among models and is unaffected by the sample size. Its calculation equation is as follows:

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_p} \quad (18)$$

where $\bar{x}_1$ and $\bar{x}_2$ represent the RMSE means of the two models respectively, and s_p represents the pooled standard deviation.

For the monthly and daily mean SST prediction tasks, we calculated Cohen's d values for our CNN-AT Transformer model compared to several baseline models, and provided the corresponding effect sizes. The results are shown in Table 13.

As shown in Table 13, in the monthly mean SST prediction, the Cohen's d value of CNN-AT Transformer compared to other baseline models ranges from 0.36 to 6.37, with an effect size between medium and large, indicating that CNN-AT Transformer model has significant advantages in medium- to long-term SST prediction. In the daily mean SST prediction, the Cohen's d value range from 0.56 to 10.99, also indicating large effect sizes. These results further validate the CNN-AT Transformer's strong capability in modeling short-term SST variability.

Table 13

The Cohen's d value and effect size of our model compared to the baseline model on the RMSE metric.

Model	Monthly SST Prediction		Daily SST Prediction	
	Cohen's d	Effect size	Cohen's d	Effect size
LSTM	6.37	Large	10.99	Large
ConVLSTM	5.23	Large	6.49	Large
MIM	4.75	Large	5.53	Large
PredRNN	2.63	Large	1.33	Large
CNN	0.58	Medium	1.94	Large
Transformer	1.34	Large	0.83	Large
TransDtSt-Part	0.47	Medium	2.02	Large
TemproNet	0.36	Medium	0.56	Medium

Table 14

The confidence intervals of RMSE scores for several models.

Model	RMSE Confidence Interval	
	Monthly Mean SST	Daily Mean SST
CNN	[0.6138, 0.6614]	[0.7192, 0.7750]
LSTM	[0.8933, 0.9603]	[1.1547, 1.2443]
ConVLSTM	[0.8385, 0.9017]	[0.9381, 1.0109]
PredRNN	[0.7124, 0.7676]	[0.6879, 0.7413]
MIM	[0.8128, 0.8794]	[0.8919, 0.9611]
Transformer	[0.6487, 0.7017]	[0.6661, 0.7169]
TemproNet	[0.6025, 0.6501]	[0.6528, 0.7028]
TransDtSt-Part	[0.6078, 0.6556]	[0.7227, 0.7797]
CNN-AT Transformer	[0.5855, 0.6313]	[0.6250, 0.6750]

To demonstrate the stability and reliability of the several SST prediction models, we evaluated the prediction error ranges using 95 % confidence intervals (CIs). Specifically, we calculated the confidence intervals for the RMSE score based on the mean and standard deviation of the samples, combined with the *t*-distribution, to reflect the error fluctuation range of the model in multiple replicates. Table 14 shows the CIs for the RMSE scores of several models for monthly and daily SST predictions.

As shown in Table 14, the RMSE CI of the CNN-AT Transformer is [0.5855, 0.6313] in the monthly mean SST prediction task, which is significantly better than that of other baseline models. In the daily mean SST prediction task, the RMSE CI of CNN-AT Transformer is [0.6250, 0.6750], which is also significantly better than other baseline models. In summary, our CNN-AT Transformer not only has a significant advantage in the average prediction error but also has a narrower CI range and is closer to the ideal value, reflecting the high stability and reliability of our model at different time scales.

4.5.3. Model deployment

Our SST prediction model has high prediction accuracy. However, when it is deployed for application, it will face problems in real-time prediction and integration with existing SST tools. The following will provide a brief discussion.

As can be seen from the model complexity analysis in Section 4.4.4, our CNN-AT transformer model has only 1.32M parameters and occupies a relatively small amount of memory. During the prediction process, its execution time is approximately 9.0s. Therefore, its execution speed is quite fast and can meet the requirements of an ordinary real-time forecasting system. For systems with extremely high real-time requirements environments or to expand deployment capabilities in an extreme resource environments, the model can be further simplified through two lightweight improvement schemes to improve real-time performance. These two schemes are Progressive Patch Pruning and Dynamic Sparse Attention. Through the above strategies, the real-time performance of the model can be further improved to meet the requirements of real-time business systems.

Our prediction model is written in Python. It has a small storage space requirement, consumes relatively little memory during operation,

and features fast execution speed and easy deployment. However, the model training takes a relatively long time. To reduce the occupation of online resources, we can adopt an offline training and online prediction approach. For model usage, we can deploy the model using a cloud-native architecture and provide services through a RESTful API interface. Specifically, it can be designed as a microservice module, encapsulated in a Docker container, and implemented with elastic scaling through a Kubernetes cluster to meet the performance requirements for real-time prediction. The model can also be deployed on cloud platforms such as Alibaba Cloud and Baidu Cloud. Access permissions can be managed through an API gateway, and standardized JSON-formatted input and output are provided. Meanwhile, the existing model can be expanded. By developing an adaptation layer to handle data format conversion, Input/Output (IO) interfaces for commonly used meteorological formats such as NetCDF and HDF5 can be supported, enabling integration with the existing forecasting system. In terms of performance optimization, strategies like model quantization, caching mechanism, and asynchronous batch processing can be adopted. For high-frequency requirements, an edge computing deployment solution can be considered.

5. Conclusions

To address the limitations of existing data-driven SST prediction methods, we propose a CNN-AT Transformer model that integrates CNN-Attention mechanisms with an improved Transformer architecture. The model is based on atmospheric and SST data. It employs CNN combined with attention mechanisms to extract and fuse shallow semantic features, thereby enhancing the ability to capture long-range dependencies and teleconnection spatiotemporal patterns. On this basis, we embed convolutional layers into the sub-layers of the Transformer model to extract and fuse local feature information. This solves the boundary artifact in image patch predictions of the traditional Transformer model. Meanwhile, we adopt a method of gradually reducing the patch size in stages to reduce the computational complexity. Ultimately, we efficiently achieve the prediction of SSTs. Extensive experiments conducted on NOAA ERSST V2 and V5 datasets, NCEP/DOE Reanalysis II dataset, and CMIP5/CMIP6 datasets demonstrate that our CNN-AT Transformer model significantly outperforms baseline methods in prediction accuracy, and has good computational efficiency, showing promising application potential.

Although the CNN-AT Transformer model has achieved significant improvements in prediction accuracy and computational efficiency, several limitations remain. Firstly, in long-term SST prediction tasks, the model's accuracy declines with increasing prediction steps due to error accumulation in multi-step predictions and the lack of explicit constraints on ocean physical processes such as ocean currents and thermohaline circulation. In short-term SST prediction scenarios, the model's responsiveness to local abrupt signals is slightly inferior to recursive-based models like PredRNN. Secondly, the model exhibits sensitivity to data quality and coverage. In high-latitude regions like the Southern Ocean, where data is sparse, the prediction errors are relatively high. Thirdly, the model currently primarily integrates SST and sea level pressure data but does not consider the influence of atmospheric circulation, solar radiation, El Niño, and La Niña phenomena on SST prediction, limiting its capability for multi-factor coupled modeling. Additionally, the model still has relatively large parameter sizes compared to traditional methods, constraining its potential for real-time deployment on edge devices like buoys and coastal stations.

To overcome these issues, future work will focus on three aspects: (1) Integrating physical oceanography knowledge (e.g., numerical ocean models or thermodynamic equations) to develop hybrid prediction models that enhance the physical consistency and predictive capabilities of SST forecasting. (2) Incorporating multi-source observational data (e.g., satellite remote sensing, Argo floats) to improve model generalization. Integrating factors such as atmospheric circulation, solar radiation, and El Niño-Southern Oscillation (ENSO) phenomena will enable more ro-

bust and reliable SST predictions while enhancing model interpretability and improve the ability to forecast extreme temperature events. (3) Designing lightweight strategies, including adaptive attention mechanisms and knowledge distillation, to further optimize model efficiency, scalability, and deployment on edge devices.

CRedit authorship contribution statement

Baiyou Qiao: Writing – review & editing, Project administration, Methodology, Investigation, Conceptualization; **Yanze Yang:** Writing – original draft, Resources, Methodology, Data curation; **Zhong Tang:** Writing – original draft, Methodology, Data curation; **Donghong Han:** Writing – review & editing, Resources, Investigation; **Gang Wu:** Writing – review & editing, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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